



Haramaya University

Department of Management

Statistics for Management I

Lecture Slide

(Mgmt 2072)



Chapter 1

Introduction



Chapter Outline

1. Definition of Statistics
2. Classification of Statistics
3. Application Areas of Statistics
4. Why We Study Statistics?
5. Data & Variable
6. Scales of Measurement

1. Definition of Statistics

- In general sense, statistics are the **collection of facts (figures)**.
- This meaning of the word is widely used when reference is made to facts and figures on sales, employment or unemployment, accident, weather, death, education, etc.
- In this sense the word Statistics serves simply as data.

Cont'd

- Specifically, statistics is the science that deals with the methods of data collection, organization, presentation, analysis and interpretation of data.
- It refers the subject area that is concerned with extracting relevant information from available data with the aim to make sound decisions.
- According to this meaning, statistics is concerned with the development and application of methods and techniques for collecting, organizing, presenting, analyzing and interpreting statistical data.

2. Classification of Statistics

- Statistics is majorly classified as:
Descriptive Statistics & Inferential Statistics
- **Descriptive Statistics** refers to the procedures used to organize and summarize masses of data.
- It is concerned with describing or summarizing the most important features of the data.
- It deals only with the characteristics of the collected data without going beyond it.

Cont'd

- The methodology of descriptive statistics includes the methods of organizing (classification, tabulation, frequency distributions) and presenting (graphical & diagrammatic presentation) data and calculations of certain indicators of data like measures of central tendency and measures of variation which summarize some important features of the data.

Cont'd

- **Inferential Statistics** includes the methods used to find out something about a population based on the sample.
- It is concerned with **drawing statistically valid conclusions** about the characteristics of the population based on information obtained from sample.
- In this form of statistical analysis, descriptive statistics is linked with probability theory in order to generalize the results of the sample to the population.

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- Performing hypothesis testing, determining relationships between variables & making predictions are inferential statistics.

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Examples: Classify the following statements as descriptive and inferential statistics:

- The average age of the students in this class is 21 years.
- At least 5% of the killings reported last year in city X were due to tourists.
- Of the students enrolled in Haramaya University in this year 74% are male and 26% are female.

Cont'd (Examples)

- The chance of winning the Ethiopian National Lottery in any day is 1 out of 167000.
- The demand for automobiles may decline next year in Europe.
- It has been continuously raining in Harar from Monday to Friday. It will continue to rain in the weekend.

3. Application Areas of Statistics

- **In Scientific Research:** Statistics plays an important role in the collection of data through efficiently designed experiments, in testing hypotheses and estimation of unknown parameters, and in interpretation of results.
- **In Industry:** Statistical techniques are used to improve and maintain the quality of manufactured goods at a desired level.

Cont'd

- **In Business:** Statistical methods are employed to forecast future demand for goods, to plan for production, and to evolve efficient management techniques to maximize profit.
- **In Medicine:** Principles of design of experiments are used in screening of drugs and in clinical trials. The information supplied by a large number of biochemical and other tests is statistically assessed for diagnosis and prognosis of disease.

Cont'd

- **In Literature:** Statistical methods are used in quantifying an author's style, which is useful in settling cases of disputed authorship.
- **In Archeology:** Quantitative assessment of similarity between objects has provided a method of placing ancient artifacts in a chronological order.

Cont'd

- **In Courts of Law:** Statistical evidence in the form of probability of occurrence of certain events is used to supplement the traditional oral and circumstantial evidence in judging cases.
- **In Detective Work:** Statistics helps in analyzing bits and pieces of information, which individually may appear to be unrelated or even inconsistent, to see an underlying pattern.

4. Why We Study Statistics?

1. Data are everywhere
2. Statistical techniques are used to **make many decisions** that affect our lives
3. No matter what your career is, you will make professional decisions that involve data. An understanding of statistical methods will help you **make these decisions effectively**.

5. Data & Variable

- Data are simply **observed facts & figures**.
- Variable is any phenomena or an attribute that can assume different values.
- The most important distinguishing feature of a variable is that it varies; that is, it can take on different values.
- Based on the values that variables assume, variables can be classified as:

Cont'd

1. **Qualitative variables:** A qualitative variable has values that are intrinsically non-numerical (categorical).
 - Qualitative data can be separated into two subgroups:
 - **Dichotomic** (if it takes the form of a word with two options (gender - male or female))
 - **Polynomic** (if it takes the form of a word with more than two options (education - primary school, secondary school & university)).

Cont'd

2. **Quantitative variables:** A quantitative variable has values that are intrinsically numerical.

Example: Height, Family size, Weight, etc.

- Quantitative variables can be further categorized into two subgroups:
 1. Discrete Variables
 2. Continuous Variables

Cont'd

1. **Discrete variables:** take whole number values and consists of distinct recognizable individual elements that can be counted.
 - It is a variable that assumes a finite or countable number of possible values. These values are obtained by counting (0, 1, 2, ...).
 - **Example:** Family size, Number of children in a family, number of cars at the traffic light.

Cont'd

2. **Continuous variables:** take any value including decimals.

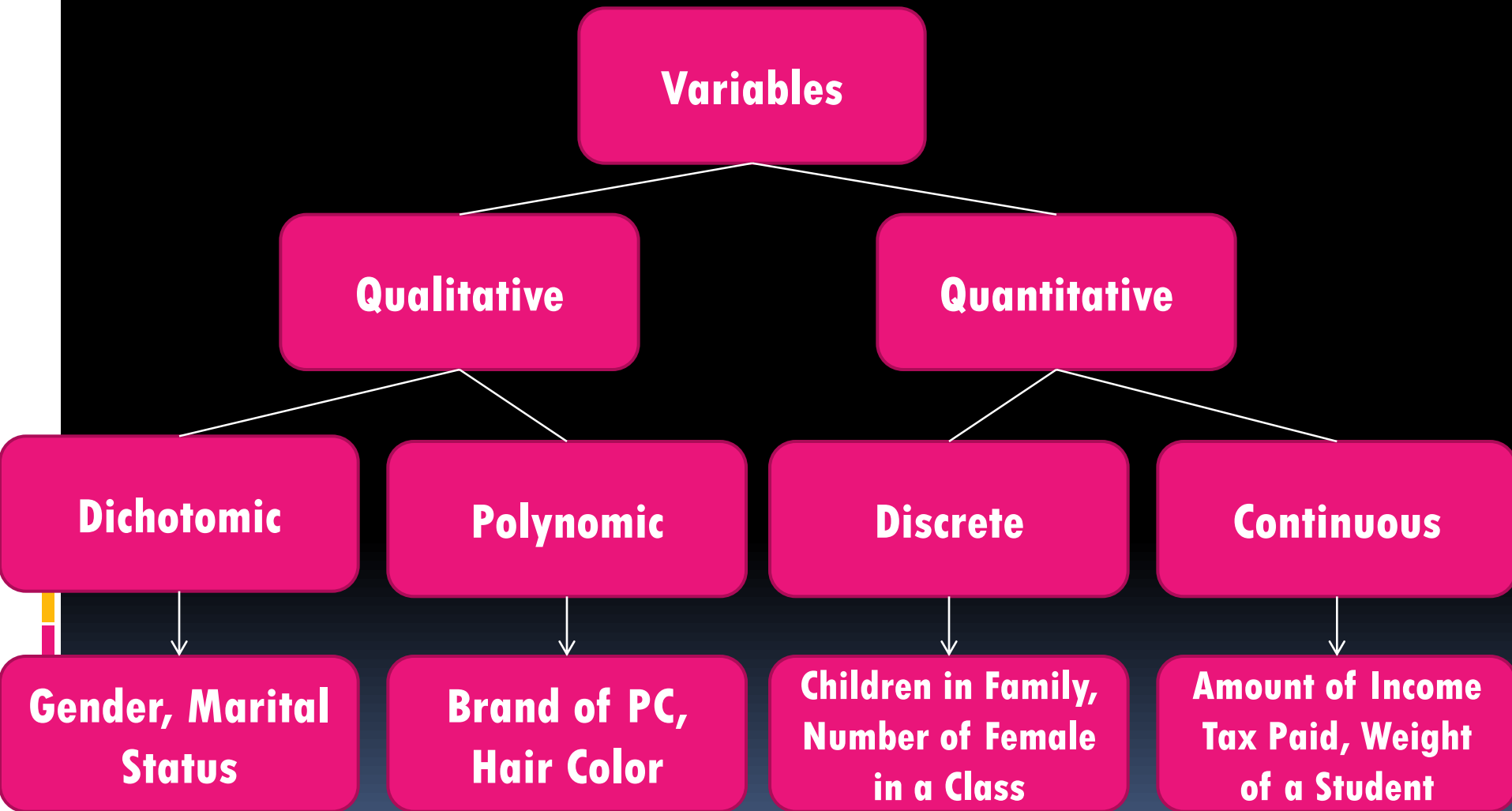
- Such a variable can theoretically assume an infinite number of possible values.
- These values are obtained by measuring.
- **Example:** Height, Weight, Time, Temperature, etc.

Cont'd

Exercise: Classify each of the following as qualitative and quantitative and if it is quantitative classify as *discrete* and *continuous*:

- ✓ Color of automobiles in a dealer's show room.
- ✓ Number of seats in a movie theater.
- ✓ Classification of patients based on nursing care needed (complete, partial or safer).
- ✓ Number of tomatoes on each plant on a field.
- ✓ Weight of newly born babies.

Summary (Types of Variables)



6. Scales of Measurement

- Different measurement scales allow for different levels of exactness, depending upon the characteristics of the variables being measured.
- The **four types of scales** available in statistical analysis are:

Cont'd

1. **Nominal Scales** of variables are those qualitative variables which show category of individuals.
 - They reflect classification in to categories (name of groups) where there is no particular order or quantitative difference to the labels.
 - **Example:** Gender (Male, Female), Political Affiliation (Labor, Conservative, Liberal), Ethnicity (White, Black, Asian, Other), etc.

Cont'd

2. **Ordinal Scales** of variables are also those qualitative variables whose values can be ordered and ranked. **Ranking and counting** are the only mathematical operations to be done on the values of the variables.
 - But there is no precise difference between the values (categories) of the variable.
 - **Example:** Academic Rank (BSc, MSc, PhD), Grade Scores (A, B, C, D, F), Strength (Very Weak, Weak, Strong, Very Strong), Health Status (Very Sick, Sick, Cured), Economic Status (Lower Class, Middle Class, Higher Class), etc.

Cont'd

3. **Interval Scales** of variables are those quantitative variables when the value of the variables is zero it does not show absence of the characteristics **i.e. there is no true zero.**

- Zero indicates lower than empty. For example, for temperature measured in degrees Celsius, the difference between 5°C and 10°C is treated the same as the difference between 10°C and 15°C. However, we cannot say that 20°C is twice as hot as 10°C, **i.e. the ratio between two different values has no quantitative meaning.**
- This is because there is no absolute zero on the Celsius scale; 0°C not imply 'no heat'.

Cont'd

4. **Ratio Scales** of variables are those quantitative variables when the values of the variables are zero, it shows absence of the characteristics.
- All mathematical operations are allowed to be operated on the values of the variables.
 - For instance, a zero unemployment rate implies zero unemployment. Thus, we can also legitimately say an unemployment rate of 20% is twice a rate of 10%.



Chapter 2

Methods of Data Collection & Presentation



Chapter Outline

1. Types of Data
2. Methods of Data Collection
3. Data Organization
4. Methods of Data Presentation

1. Types of Data

- Research results or findings reveal information that are obviously an output of properly and carefully collected relevant data, after they are being analyzed through legitimate data analysis instruments.
- So, data are always a base (or an input) for research.
- This implies that the quality of our study is heavily dependent on the quality of our data.

Cont'd

- Data can be collected from different sources which are generally grouped under two major categories, namely, **primary & secondary sources of data**.
- Primary data is the one which is collected by the investigator himself for the purpose of a specific inquiry or study.

Cont'd

- These data are those data collected for the first time either through direct observation or by enquiring individuals under the direct supervision and instruction of the researcher.
- When an investigator uses the data which has already been collected by others, such data is called *secondary data*.

Cont'd

- Based on the role of time, data can be classified as *cross-sectional & time series*.
- **Cross-sectional data** is a set of observations taken at a point of time.
- **Time series data** is a set of observations collected for a sequence of time usually at equal intervals.

2. Methods of Data Collection

There are three major primary methods of data collection:

1. Survey Method
2. Experimentation
3. Direct Observation

Cont'd

- Secondary data should be used with utmost care.
- The investigator, before using these data, must observe that they possess the following characteristics: reliability of data, suitability of data & adequacy of data.

3. Data Organization

- It is almost impossible for management to deal with all the collected data in the raw form as it is in a haphazard and unsystematic form.
- In order to describe situations and make inferences about the population even to describe the sample, the data must be organized into some meaningful way.

Cont'd

- Before further analysis, the collected data should be **edited** for completeness, consistency, accuracy and homogeneity.
- The next important step towards organizing data is **classification**.
- Classification is the separation of items according to similar characteristics and grouping them into various groups.

Cont'd

Tabulation of Data:

- A table is a systematic arrangement of data in **rows & columns**, which is easy to understand and makes data fit for further analysis and drawing conclusions.
- A statistical table, in general, should have the following parts:

Cont'd

- **Table Number:** Every table should be identified by a number. It facilitates easy reference.
- **Title:** There should be a title at the top of every statistical table. The title should be clear, concise and adequate.
- **Stub:** It is a title given to each row.
- **Caption:** The caption labels the data presented in a column of the table. There may be sub-captions in each caption.

Cont'd

- **Body:** The body of the table is the most important part. The information given in the rows and columns forms the body of the table.
- **Footnote:** Any explanatory notes concerning the table itself, placed directly beneath the table, is called 'footnote'. The main purpose of footnote is to clarify some of the specific items given in the table or to explain the ambiguities, omissions, if any, about the data shown in the table.

Cont'd

- **Source Note:** If the data is collected from secondary sources, a source note is given to disclose the sources from which the data is collected.
- **Example:** Consider the following format:

Cont'd (Example)

Table 2.1: Title of your table

	Caption 1	Caption 2	Caption 3
Stub 1	15	65	3.5
Stub 2	22	88	6.3
Stub 3	a*	78	5.3

- *No caption is found for stub 3

Cont'd

● Frequency Distributions:

- The most convenient way of organizing numerical data is to construct a frequency distribution.
- Frequency distribution is the organization of raw data in table form, using **classes & frequencies**.
- Here the term '**class**' is a description of a group of similar numbers in a data set while '**frequency**' is the number of times a variable value is repeated.

Cont'd

- There are three types of frequency distributions: **categorical, ungrouped & grouped frequency distributions.**
- **Categorical Frequency Distribution:** the data is qualitative i.e. either nominal or ordinal.
- Each category of the variable represents a single class and the number of times each category repeats represents the frequency of that class (category).

Cont'd

- **Example:** The blood type of 22 students is given below. Construct categorical frequency distribution.

A B B AB O A O O B AB B

A B B O A O AB A O O AB

Solution

Cont'd

- **Ungrouped Frequency Distribution:** A frequency distribution of numerical data (quantitative) in which each value of a variable represents a single class.
- The values of the variable are not grouped and the number of times each value repeats represents the frequency of that class.

Cont'd

- **Example:** Number of children for 21 families is:

2 3 5 4 3 3 2 3 1 0 4 3 2 2 1 1 1 4 2 2 2

- Required: Construct ungrouped frequency distribution.

Solution

Cont'd

- **Grouped (Continuous) Frequency Distribution:** A frequency of numerical data in which several values of a variable are grouped into one class.
- The number of observations belonging to one class is the frequency of the class.
 - ▣ **Example:** Consider age group and number of persons:

Cont'd (Example)

Class Limits	Class Boundaries	Frequency
1 – 25	0.5 – 25.5	20
26 – 50	25.5 – 50.5	15
51 – 75	50.5 – 75.5	25
76 – 100	75.5 – 100.5	10
Total		70

Cont'd

- **Class Limits:** the lowest and highest values that can be included in a class are called class limits.
- The lowest values are called **lower class limits** and the highest values are called **upper class limits**.
- **Class Boundaries:** are class limits when there is no gap between the UCL of the first class and the LCL of the second class.
- The lowest values are called **lower class boundaries** and the highest values are called **upper class boundaries**.

Cont'd

- **Class Width:** the difference between UCB and LCB of a class.
- It is also the difference between the lower limits of two consecutive classes or it is the difference between upper limits of two consecutive classes.
- **Class Mark:** is the half way between the class limits or the class boundaries

$$cm_i = \frac{LCL_i + UCL_i}{2} \text{ or } \frac{LCB_i + UCB_i}{2}$$

Cont'd

- **Relative Frequency:** the relative frequency distribution can be formed by dividing the **frequency in each class (absolute frequency)** of the frequency distribution by the total number of observations.
- It can be converted in to a **percentage frequency distribution** by simply multiplying each relative frequency by 100.

Cont'd

Class Limits	Class Boundaries	Absolute Frequency	Relative Frequency	%tage Frequency
1 – 25	0.5 – 25.5	20		
26 – 50	25.5 – 50.5	15		
51 – 75	50.5 – 75.5	25		
76 – 100	75.5 – 100.5	10		
Total		70		

Cont'd

- **Cumulative Frequency:** A cumulative frequency distribution displays the total number of observations above or below a certain value.
- When the interest of the investigator focuses on the number of items below a specified value, it is known as **less than cumulative frequency distribution** and when the focus is on the number of items above a specified value, it is known as **more than cumulative frequency distribution**.

Cont'd

Class Limits	Class Boundaries	Frequency	LCF	MCF
1 – 25	0.5 – 25.5	20		
26 – 50	25.5 – 50.5	15		
51 – 75	50.5 – 75.5	25		
76 – 100	75.5 – 100.5	10		
Total		70		

Cont'd

Steps for the Construction of Grouped Frequency Distribution

1. Arrange the data in an array form (increasing or decreasing order).
2. Find the unit of measurement (u). ' u ' is the smallest difference between any two distinct values of the data.
3. Find the Range(R). ' R ' is the difference between the largest and the smallest values of the variable.

$$R = \text{Max} - \text{Min}$$

Cont'd (Steps)

4. Determine the number of classes (k) using Sturge's rule.

$$k = 1 + 3.322 \log N$$

where 'N' is the total number of observations.

5. Specify the class width (w).

$$w = R/k$$

6. Put the smallest value of the data set as the LCL of the first class.

Cont'd

$$LCL_1 = x$$

$$LCL_i = LCL_{i-1} + w \text{ for } i = 2, 3, \dots, k \quad \&$$

$$UCL_i = LCL_i + (w - u) \text{ for } i = 2, 3, \dots, k.$$

7. Generate the class boundaries.

$$LCB_i = LCL_i - u/2 \text{ and } UCB_i = UCL_i + u/2 \text{ for } i = 2, 3, \dots, k.$$

Cont'd

Example: Mark of 50 students out of 40.

16 21 26 24 11 17 25 26 13 27 24 26 3
27 23 24 15 22 22 12 22 29 18 22 28 25
7 17 22 28 19 23 23 22 3 19 13 31 23 28
24 9 20 33 30 23 20 8 21 24

Required: Construct grouped frequency distribution for the given data set.

Solution

Cont'd (Solution)

Class Limits	Class Boundaries	Class Mark	Frequency
3 – 7	2.5 – 7.5	5	3
8 – 12	7.5 – 12.5	10	4
13 – 17	12.5 – 17.5	15	6
18 – 22	17.5 – 22.5	20	13
23 – 27	22.5 – 27.5	25	17
28 – 32	27.5 – 32.5	30	6
33 – 37	32.5 – 37.5	35	1
Total			50

4. Methods of Data Presentation

- Such methods provide summary information about a data set and may be used to conduct exploratory data analyses.
- Showing data in the form of a graph can make complex and confusing information appear more simple & straightforward.

A. Bar Chart

- It is the simplest and most commonly used diagrammatic representation of a frequency distribution.
- It is the most common presentation for **nominal, categorical or discrete data**.
- It uses a series of separated and equally spaced bars.
- The heights of the bars represent the frequency or relative frequency of the classes.

Cont'd

- But the width of the bars has no meaning; however, all the bars should be the same width to avoid distortion.
- And also the bars are separated by constant distance.

Simple Bar Chart

- This is a diagram in which categories of a variable are marked on the X-axis and the frequencies of the categories are marked on the Y-axis.
- It is applicable for discrete variables, that is, for data given according to some period, places & timings.

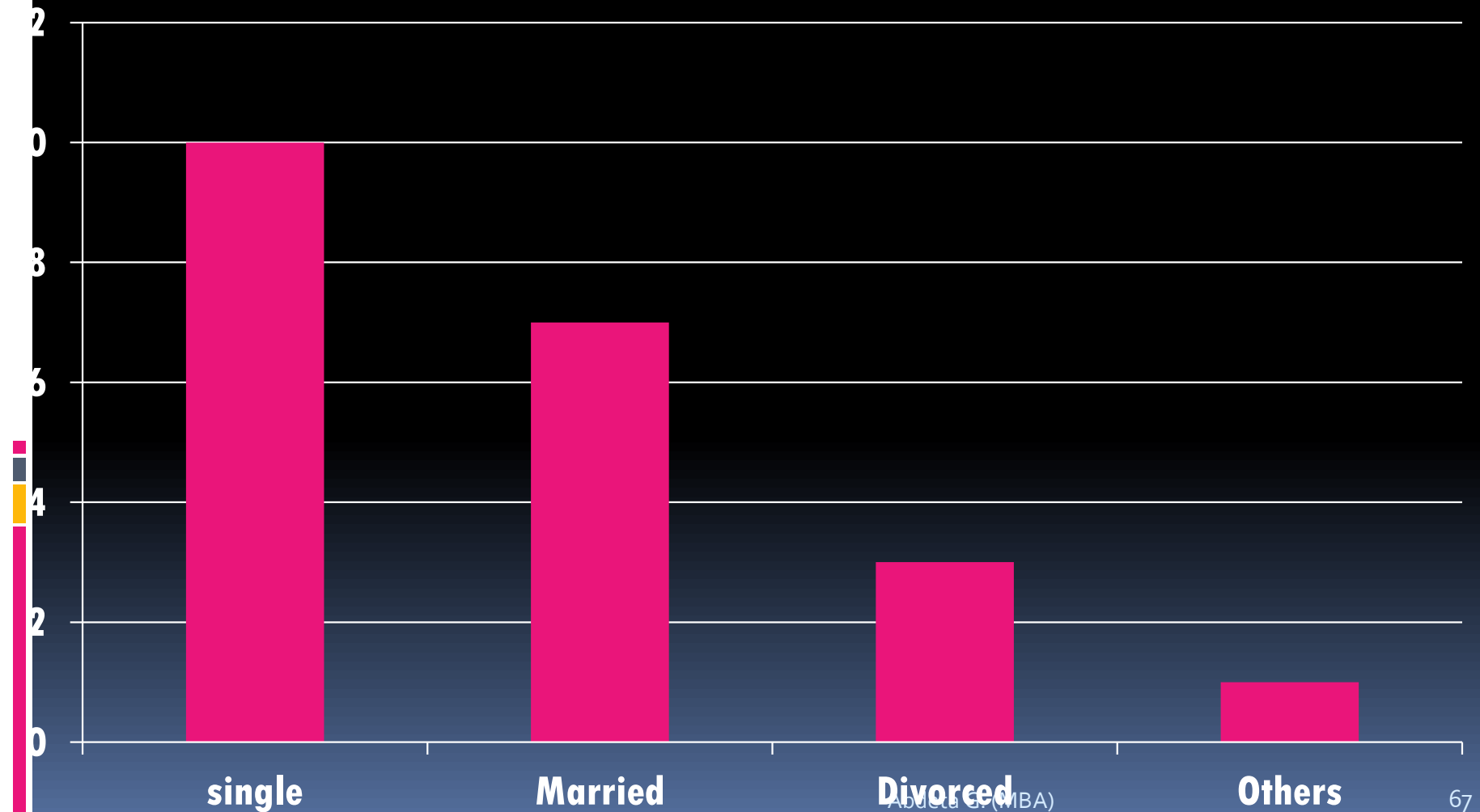
Cont'd

- **Example:** Construct simple bar chart for the following data.

Marital Status	Number of Individuals
Single	10
Married	7
Divorced	3
Others	1
Total	21

Cont'd

Simple Bar Chart



Component Bar Chart

- This is used when there is a desire to show a total or aggregate is divided into its component parts.
- The bars represent total value of a variable with each total broken into its component parts and different colors are used for identification.
- For making meaningful comparisons, the components of the attributes can be reduced to percentages. This sort of component bar chart is known as **percentage bar chart**.

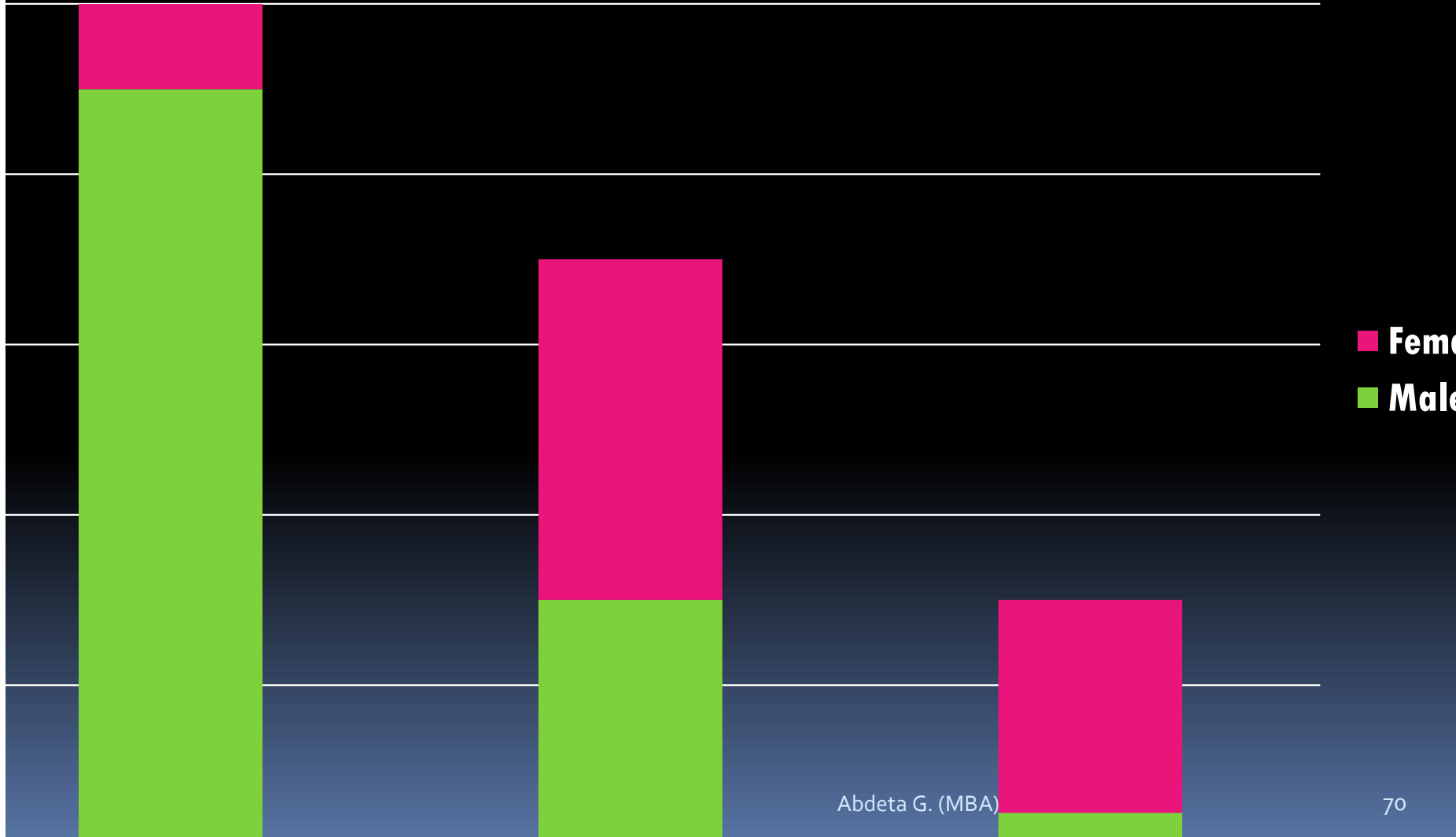
Cont'd

- **Example:** Construct bar chart for the following data.

Marital Status	Male	Female	Total
Single	90	10	100
Married	30	40	70
Others	5	25	30

Component Bar Chart

Cont'd



Multiple Bar Chart

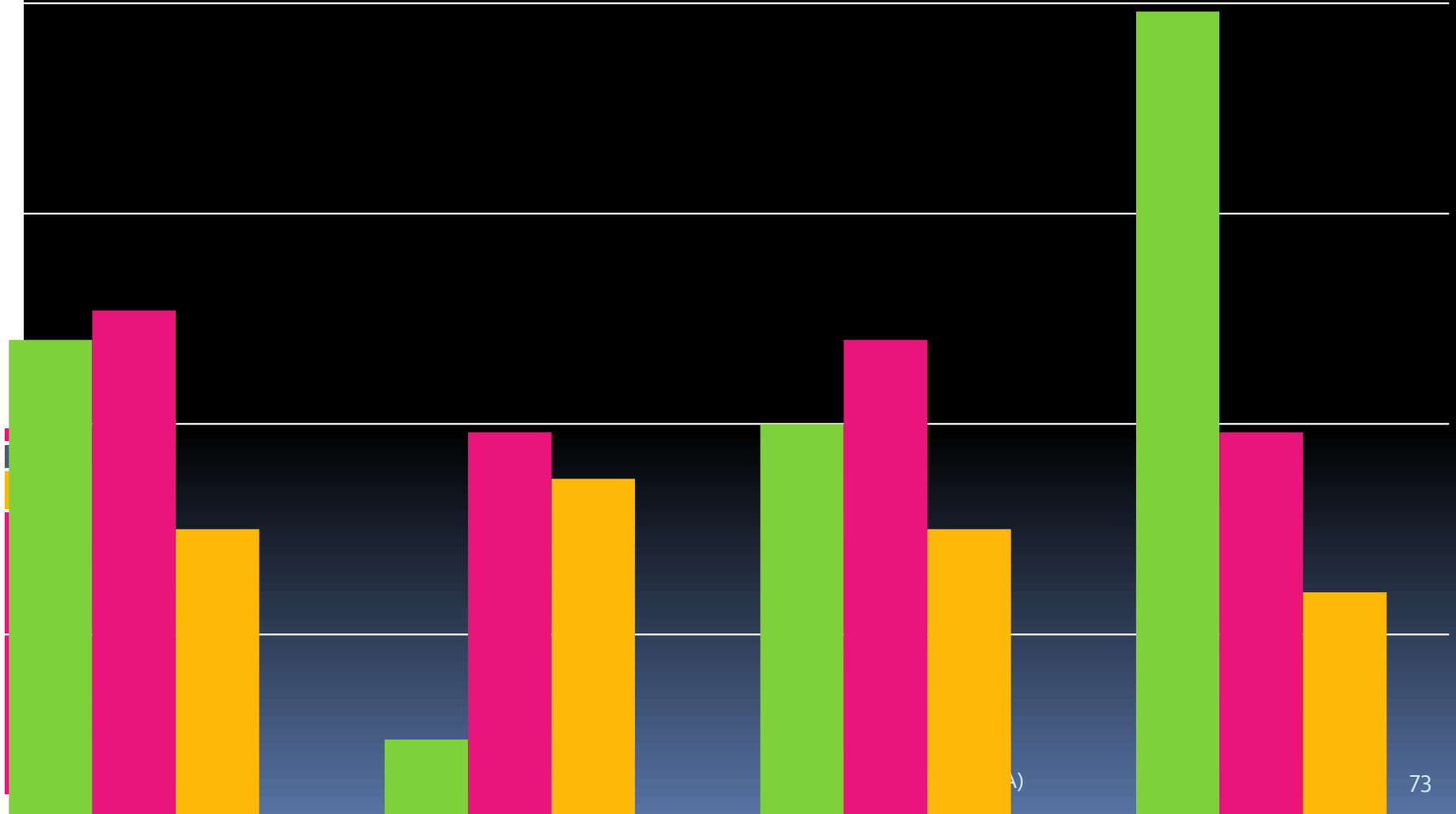
- This is used to display data on more than one variable.
- In the multiple bars diagram two or more sets of inter-related data are interpreted.
- **Example:** Consider the following table which shows the export of some items for a given country and show it using bar chart.

Cont'd

Year	Coffee	Butter	Sugar
1997	120	127	75
1998	25	98	87
1999	100	120	75
2000	198	98	60

Multiple Bar Chart

Cont'd



B. Pie Chart

- **Pie chart is popularly used in practice to show percentage break down of data.**
- **It is a circle representing a set of data by dividing the circle into sectors proportional to the number of items in the categories or it is a circle representing the total, cut into slices in proportional to the size of the parts that make up the total.**

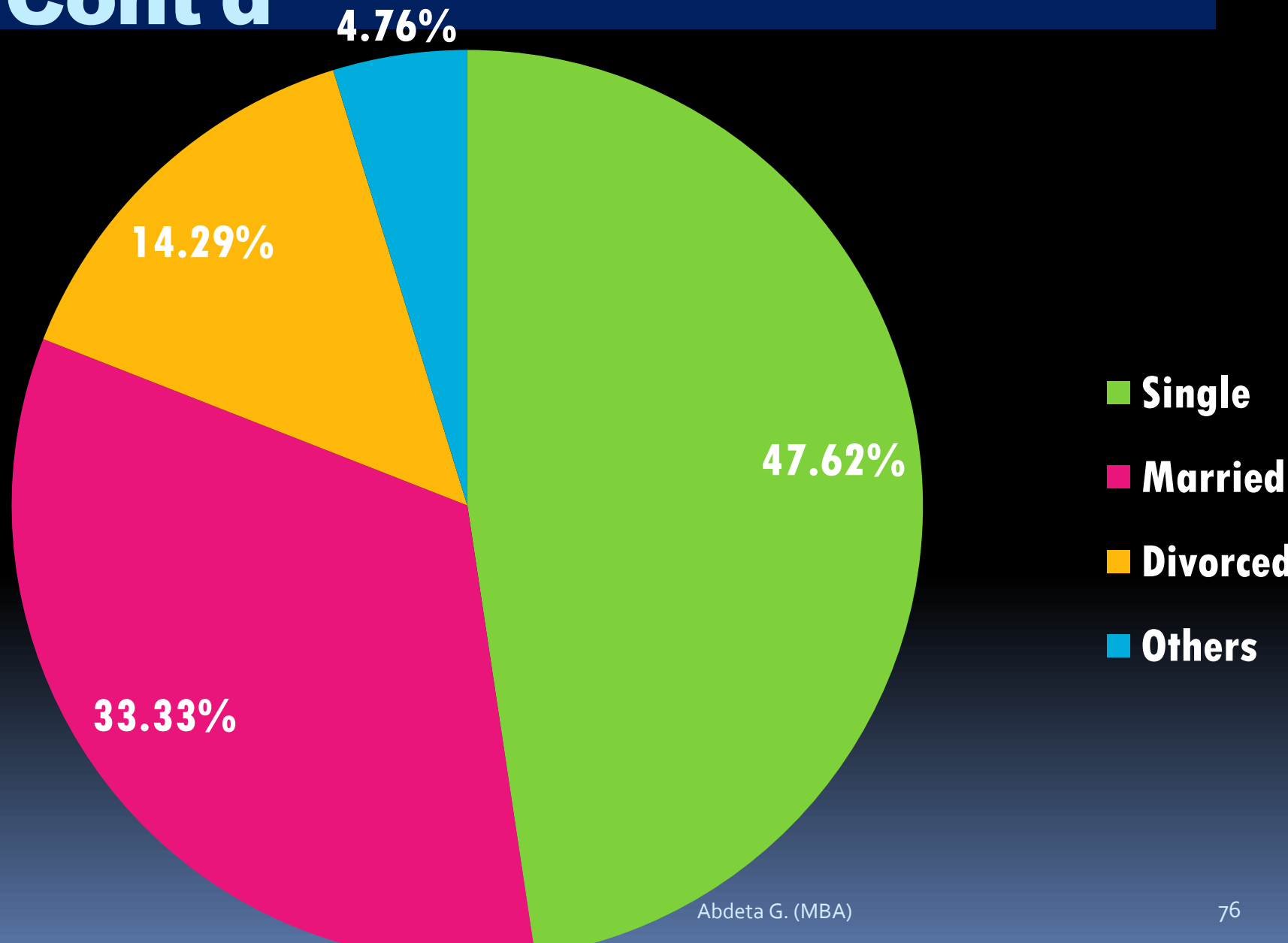
Cont'd

- **Example:** Construct a pie chart for the following data.

Marital Status	Number of Individuals	Percentage	Degree
Single	10	47.62	171.43
Married	7	33.33	119.99
Divorced	3	14.29	51.44
Others	1	4.76	17.14
Total	21	100	360

Pie Chart

Cont'd



C. Histogram

- Histogram is the most common graphical presentation of a frequency distribution for *numerical data*.
- It uses a series of adjacent bars in which the width of each bar represents the class width and the heights represent the frequency or relative frequency of the class.
- It is used **for grouped data** in which the class boundaries are marked on the X-axis and the frequencies are marked along the Y-axis.

Cont'd

- **Example:** In the following data, the heights of 45 female students at Haramaya University are recorded to the nearest inch. Construct a histogram for it.

67 67 64 64 74 61 68 71 69 61 65 64 62 63 59
70 66 66 63 59 64 67 70 65 66 66 56 65 67 69
64 67 68 67 67 65 74 64 62 68 65 65 65 66 67

Cont'd

- **First we have to construct a grouped frequency distribution:**

✓ $U = 65 - 64 = \underline{1}$

✓ $R = 74 - 56 = \underline{18}$

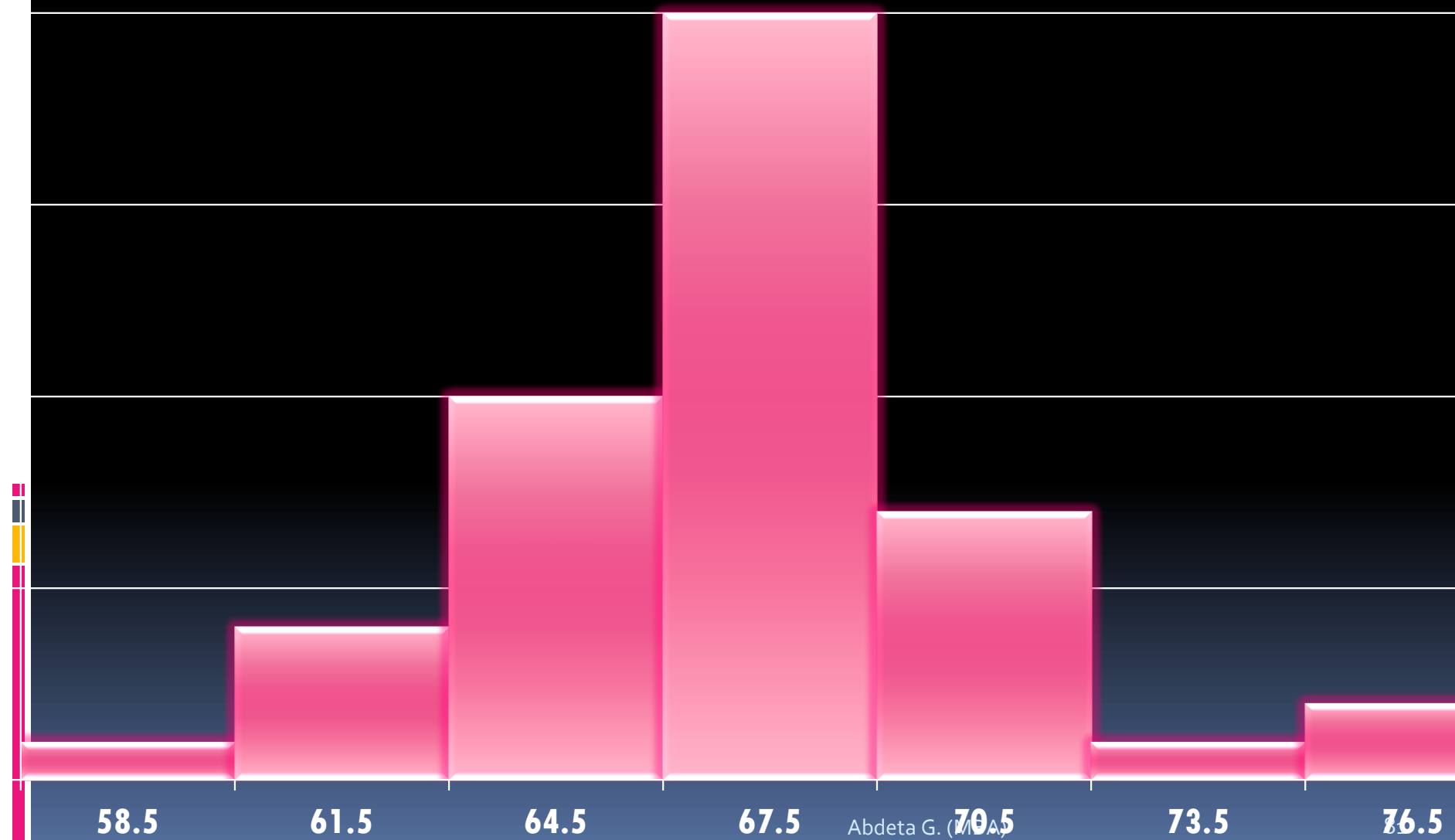
✓ $K = 1 + 3.322 \log N = 1 + 3.322 \log 45$
 $= 1 + 3.322 (1.65) = 6.50 \approx \underline{7}$

✓ $W = R/K = 18/7 = 2.57 \approx \underline{3}$

Cont'd

Class Limits	Class Boundaries	Frequency
56 – 58	55.5 – 58.5	1
59 – 61	58.5 – 61.5	4
62 – 64	61.5 – 64.5	10
65 – 67	64.5 – 67.5	20
68 – 70	67.5 – 70.5	7
71 – 73	70.5 – 73.5	1
74 – 76	73.5 – 76.5	2
Total		45

Cont'd



Abdeta G. (MBA)

D. Frequency Polygon

- **It is a graph that consists of line segments connecting the intersection of the class marks and the frequencies of a continuous frequency distribution.**
- **It can also be constructed from histogram by joining the mid-points of each bar.**

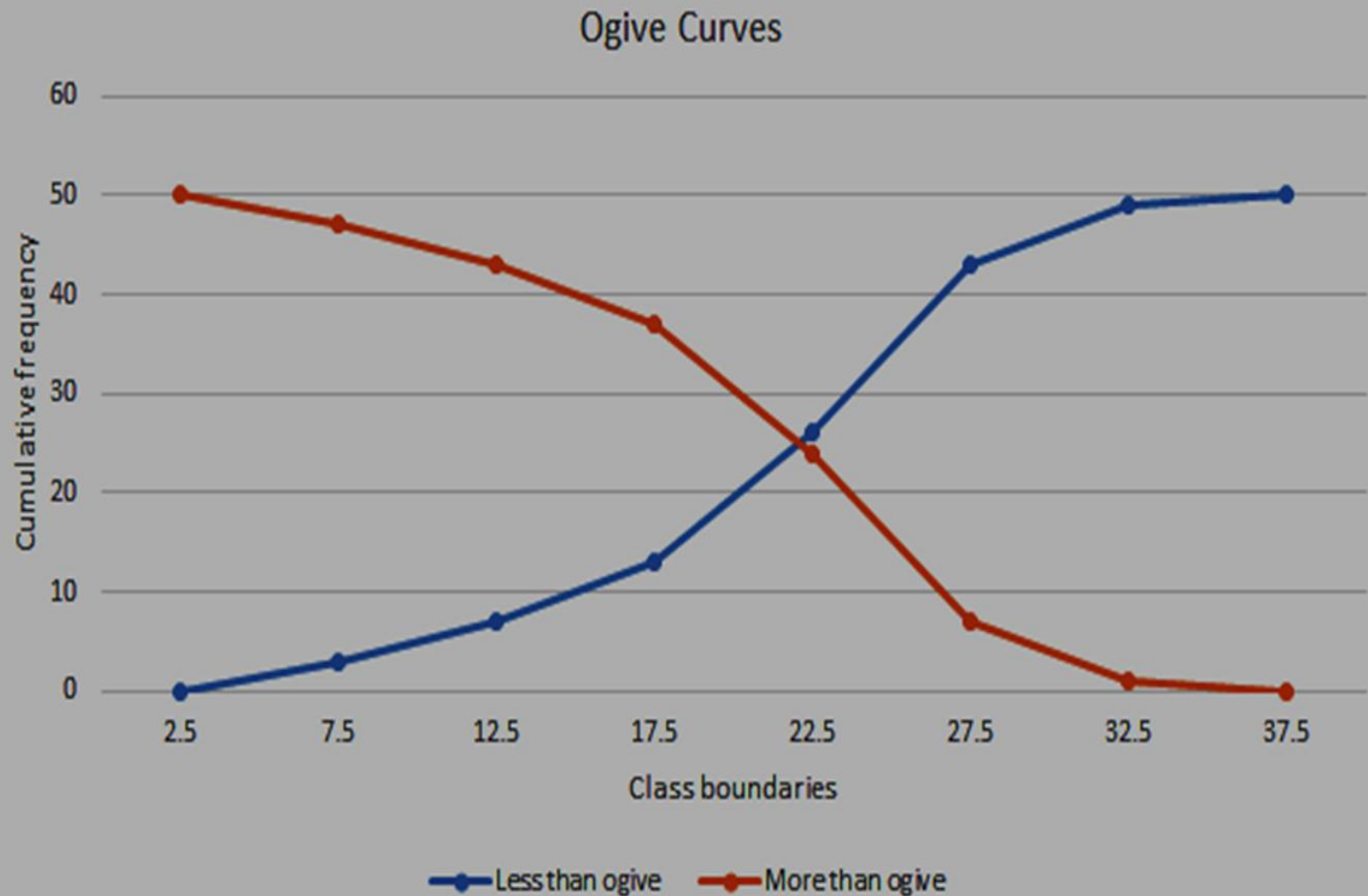
E. Ogive

- It is the curve of cumulative frequencies.
- As there are two cumulative frequency distributions, there are two ogive (pronounced as “**oh-jive**”) curves.
- These are the less than cumulative frequency which is a line graph joining the intersection points of the upper class boundaries and their corresponding less than cumulative frequencies &

Cont'd

- The more than cumulative frequency which is a line graph joining the intersection points of the lower class boundaries and their corresponding more than cumulative frequencies.
- **Example:** Consider the following ogive curves for the marks of 50 students given on the previous example.

Cont'd





Chapter 3

Measures of Central Tendency & Dispersion

1. Measures of Central Tendency

- A single value which can be considered as a typical or representative of a set of observations and around which the observations can be considered as centered is called an average (or average value or center of location).
- Since such typical values tend to lie centrally within a set of observations when arranged according to magnitudes; averages are called *measures of central tendency (MCT)*.

Objectives of MCT

Two Major Objectives:

- 1. To condense a mass of data in to one single value**
- 2. To facilitate comparison**

Desirable Properties of Good MCT

- It should be calculated based on all observations**
- It should not be affected by extreme values**
- It should be defined rigidly which means it should have a definite value**
- It should always exist**
- It should be easy to understand and calculate**
- It should be capable of further algebraic treatment**

Summation Notation

- Suppose we have variable ' x ' having successive values $x_1, x_2, \dots, x_n$.
- The sum of these values can be written as $x_1 + x_2 + \dots + x_n$.
- This can be written as using Greek letter Σ as:

$$x_1 + x_2 + \dots + x_n = \sum_{i=1}^n x_i$$

Types of MCT

- **The most frequently encountered measures of central tendencies are:**
 - 1. Mean (computed average)**
 - **Arithmetic mean (simple arithmetic mean, weighted arithmetic mean & combined mean)**
 - **Geometric mean**
 - **Harmonic mean**

Cont'd

- 2. Mode (the most frequented value)**
- 3. Positional averages**
 - Median**
 - Quantiles (quartiles, deciles & percentiles)**

1. Mean

A. Arithmetic Mean

🌿 Simple Arithmetic Mean

1. Suppose a variable x has observed values $x_1, x_2, \dots, x_n$. The simple arithmetic mean denoted by $\bar{x}$ (for sample) and μ (for population) is the sum of these observations divided by the total number of observations. Symbolically:

Cont'd

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\mu = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{\sum_{i=1}^N x_i}{N}$$

2. Suppose the values $x_1, x_2, \dots, x_n$ are accompanied by frequencies $f_1, f_2, \dots, f_n$ respectively, then the simple AM is given by:

Cont'd

- $$\bar{x} = \frac{f_1x_1 + f_2x_2 + \cdots + f_nx_n}{f_1 + f_2 + \cdots + f_n} = \frac{\sum f_i x_i}{\sum f_i}$$

3. For data in grouped frequency distribution we use the class mark instead of each observed value and simple AM is given by:

$$\bar{x} = \frac{f_1m_1 + f_2m_2 + \cdots + f_nm_n}{f_1 + f_2 + \cdots + f_n} = \frac{\sum f_i m_i}{\sum f_i}$$

Where, m_i is the class mark of the i^{th} class.

Cont'd

- **Example 1:** The heights of 7 students selected from a class are given in centimeter as 165, 160, 172, 168, 159, 170, 173. Calculate the simple AM of heights.

Soln

Cont'd

- **Example 2:** The following is the frequency distribution of marks in Stat 1011 of 46 students (out of 20). Find the mean mark of this class.

Mark (xi)	9	10	11	12	13	14	15	16	17	18	Total
No of Stdts (fi)	1	2	3	6	10	11	7	3	2	1	46
fixi	9	20	33	72	130	154	105	48	34	18	623

Soln

Cont'd

- **Example 3:** Calculate the mean amount of yield of maize from the grouped frequency distribution given below.

Yield (in kg)	No of Plots (fi)	Class mark (mi)	fimi
171 – 179	3	175	525
180 – 188	7	184	1288
189 – 197	12	193	2316
198 – 206	9	202	1818
207 – 215	4	211	844
216 – 224	4	220	880
225 – 233	1	229	229
Total	40		7900

Cont'd

■ Weighted Arithmetic Mean

- It is an arithmetic mean used when all observations in data have unequal relative importance (technically termed as weight).
- Suppose $x_1, x_2, \dots, x_n$ have weights $w_1, w_2, \dots, w_n$ respectively, then weighted arithmetic mean ($\bar{x}_w$) is given by:

$$\bar{x}_w = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum w_i x_i}{\sum w_i}$$

Cont'd

- **Example:** Semester grade point average (GPA) of a student is a good example of weighted arithmetic mean. Calculate the GPA of this student?

Course	Weights (Credit hours)	Grade (x)
Stat 281	4	B = 3
Mgmt 261	4	B = 3
Mgmt 224	3	C = 2
PADM 201	3	B = 3
Comp 201	3	C = 2

Soln

Cont'd

■ Combined Mean

- If there are k different groups (having the same unit of measurement) with mean $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$ and number of observations $n_1, n_2, \dots, n_k$ respectively, then the mean of all the groups i.e. the combined mean is given by:

$$\bar{\bar{x}} = \bar{x}_c = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_k\bar{x}_k}{n_1 + n_2 + \dots + n_k} = \frac{\sum n_i\bar{x}_i}{\sum n_i}$$

Cont'd

- **Example:** There are 49 students in a certain department. Among these 7 are seniors with average weight of 165 lbs, 9 are juniors with average weight of 160 lbs, 13 are sophomores with average weight of 152 lbs and 20 freshman with average weight of 150 lbs. Find the average weight of students in the department.

Soln

Properties of Arithmetic Mean

- If a constant ' k ' is added or subtracted from each value in a distribution, then the new mean will be:

$$\bar{x}_{new} = \bar{x}_{old} \pm k$$

- If each value of a distribution is multiplied by a constant ' k ', the new mean will be the original mean multiplied by k . That is,

$$\bar{x}_{new} = k\bar{x}_{old}$$

- Arithmetic mean is highly affected by extreme values.

Cont'd

- **Arithmetic mean can be calculated for any set of data (quantitative data), and it will be unique. We cannot calculate AM for open-ended grouped frequency distribution.**
- **It lends itself for further statistical analysis. For example, as combined mean.**
- **The algebraic sum of the deviations of each value from the arithmetic mean is zero. That is,**

$$\sum (x_i - \bar{x}) = 0$$

Cont'd

- **Example 1:** The mean age of a group of 100 students was found to be 32.02 years. Later it was discovered that age of 57 was misread as 27. Find the correct mean.

Soln

- **Example 2:** The mean weight of 150 students in certain class is 60 kg. The mean weight of boys in the class is 70 kg and that of the girls is 55 kg. Find the number of boys and girls in the class.

Soln

Cont'd

B. Geometric Mean (GM)

- The geometric mean of n -positive numbers is the n^{th} root of their product.
- The geometric mean of $x_1, x_2, \dots, x_n$ is given by the following for raw data, ungrouped and grouped frequency respectively.

$$GM = \sqrt[n]{x_1 \times x_2 \times \cdots \times x_n} = \sqrt[n]{\prod_{i=1}^n x_i}$$

Cont'd

■

$$GM = \sqrt{x_1^{f_1} \times x_2^{f_2} \times \cdots \times x_n^{f_n}} = \sqrt[n]{\prod_{i=1}^n x_i^{f_i}}$$

$$GM = \sqrt{m_1^{f_1} \times m_2^{f_2} \times \cdots \times m_n^{f_n}} = \sqrt[n]{\prod_{i=1}^n m_i^{f_i}}$$

Cont'd

- We can also use logarithms to calculate **GM**

$$GM = \sqrt[n]{x_1} \times x_2 \times \cdots \times x_n = (x_1 \times x_2 \times \cdots \times x_n)^{1/n}$$

$$\text{Log}GM = \frac{1}{n} \log(x_1 \times x_2 \times \cdots \times x_n)$$

$$\text{Log}GM = \frac{1}{n} \log(\log x_1 + \log x_2 + \cdots + \log x_n)$$

Taking the antilog of both sides we get:

$$\begin{aligned} GM &= \text{antilog} \left\{ \frac{1}{n} \log(\log x_1 + \log x_2 + \cdots + \log x_n) \right\} \\ &= \text{antilog} \left(\frac{1}{n} \sum \log x_i \right) \end{aligned}$$

Cont'd

- **Example 1:** A given epidemic was spreading at the rate of 2.5 and 3.5 in two successive days. What is its average spread rate?

Soln

- **Example 2:** The price of a commodity increased by 5% from 1989 to 1990, 10% from 1990 to 1991 and by 50% from 1991 to 1992. Find the average price increase.

Soln

Cont'd

- **Example 3:** A machine depreciated by 20% each in the first two years and by 50% in the third year. Find out the average rate of depreciation.

Soln

Cont'd

C. Harmonic Mean (HM)

- Harmonic mean is another specialized average which is useful in averaging variables expressed as rate per unit of time such as speed, number of units produced per day.
- Simple harmonic mean is the **reciprocal of the arithmetic mean** of the numbers.

Cont'd

For Raw Data:

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum \frac{1}{x_i}}$$

For Ungrouped Frequency Distribution:

$$HM = \frac{f_1 + f_2 + \dots + f_n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}} = \frac{\sum f_i}{\sum \frac{f_i}{x_i}}$$

For Grouped Frequency Distribution:

$$HM = \frac{f_1 + f_2 + \dots + f_n}{\frac{f_1}{m_1} + \frac{f_2}{m_2} + \dots + \frac{f_n}{m_n}} = \frac{\sum f_i}{\sum \frac{f_i}{m_i}}$$

Cont'd

- The weighted *HM* of '*n*' non-zero observations $x_1, x_2, \dots, x_n$ having weights $w_1, w_2, \dots, w_n$ respectively is given by:

$$HM = \frac{w_1 + w_2 + \dots + w_n}{\frac{w_1}{x_1} + \frac{w_2}{x_2} + \dots + \frac{w_n}{x_n}} = \frac{\sum w_i}{\sum \frac{w_i}{x_i}}$$

Cont'd

- **Example 1:** A driver travels for 3 days at speed of 48 km/hr for about 10 hrs, 40 km/hr for 12 hrs, 32 km/hr for 15 hrs respectively. What is the average speed of the driver in 3 days?

Soln

- **Example 2:** A driver travelled for 3 days on first days he derived for 10 hrs at speed of 48 km/hr, on the second day for 12 hrs at 45 km/hr, on third day for 15 hrs at 40 km/hr. What is the average speed?

Soln

Cont'd

- **Example 3:** A trip from town 'A' to 'B' at speed of 40km will take 1 hour. A trip back from town 'B' to 'A' at speed of 80km will take 0.5 hours. What will be the average speed of the trip?

Soln

Some Empirical Relationship among AM, GM & HM

- The GM of two numbers x_1 and x_2 is equal to the GM of their AM and HM. That is,

$$GM = \sqrt{x_1 \times x_2} = \sqrt{AM \times HM}$$

- For ' n ' positive numbers $HM \leq GM \leq AM$.

2. Mode

- The mode (modal value) of data set is the value that **occurs most frequently**.
- When two values occur with the same greatest frequency, each one is a mode and the data set is **bimodal**.
- When more than two values occur with the greatest frequency, each is a mode and the data set is said to be **multimodal**.
- When no value is repeated or all values are equally repeated, we say that there is **no mode**.

Cont'd

Example 1: Find the modes of the following data sets:

- 5 5 5 3 1 5 1 4 3 5
- 1 2 2 2 3 4 5 6 6 6 7 9
- 1 2 3 6 7 8 9 10

Soln

Cont'd

- In a frequency distribution, the mode is located in the class with highest frequency and that class is the modal class. Then the formula for mode is:

$$\hat{x} = L_{\hat{x}} + \left[\frac{f_{\hat{x}} - f_{\hat{x}-1}}{(f_{\hat{x}} - f_{\hat{x}-1}) + (f_{\hat{x}} - f_{\hat{x}+1})} \right] w$$

Where:

- $L_{\hat{x}}$ is the lower class boundary of the modal class,

Cont'd

- $f_{\hat{x}}$ is the frequency of modal class,
- $f_{\hat{x}-1}$ is the frequency of the class which immediately precedes the modal class,
- $f_{\hat{x}+1}$ is the frequency of the class which is immediate successor of the modal class &
- w is the class width of the modal class.

Cont'd

- **Example:** Use the frequency distribution of heights in the following table to find the mode of height of the 100 female students at Haramaya University and interpret the result.

Heights (interval)	Frequency (fi)
60.5 – 63.5	5
63.5 – 66.5	18
66.5 – 69.5	42
69.5 – 72.5	27
72.5 – 75.5	8

Soln

Cont'd

- Mode is not affected by extreme values and can be calculated for **open-ended classes**.
- But it often does not exist and its value may not be unique. In such case mode is **ill-defined**.



Properties of Mode

- **It is simple to calculate and easy to determine.**
- **It is not based on all observations.**
- **The mode can be used for both qualitative (such as religious preference, gender, political affiliation, etc.) & quantitative data types.**

3. Median(Positional Average)

- A median is a value which **divides set of data in to two equal parts** such that the number of observations below it is the same as the number of observations above it.
- It is the middle value when the values are arranged in order of increasing (or decreasing) magnitude.
- To find the median, first sort the values (arrange them in order), then use one of the following procedures:

Cont'd

- †. If the number of values is **odd**, the median is the number that is located in the exact middle of the list.

$$\tilde{x} = \left(\frac{n + 1}{2} \right)^{th} \text{ value}$$

- **Example:** What is the median of 180, 201, 220, 191, 219, 209 & 220

Soln

Cont'd

2. If the number of values is **even**, the median is found by computing the mean of the two middle numbers.

$$\tilde{x} = \frac{\left(\frac{n}{2}\right)^{th} \text{ value} + \left(\frac{n}{2} + 1\right)^{th} \text{ value}}{2}$$

- **Example:** What is the median of 62, 63, 64, 65, 66, 66, 68 & 78.

Soln

Cont'd

3. For grouped frequency distributions median is given by the formula:

$$\tilde{x} = L_{\tilde{x}} + \left(\frac{\frac{n}{2} - F_{\tilde{x}-1}}{f_{\tilde{x}}} \right) w$$

Where:

- $L_{\tilde{x}}$ is the lower class boundary of the median class,
- $F_{\tilde{x}-1}$ is the less than cumulative frequency just before the median class,

Cont'd

- w is the class width of the median class,
- $f_{\tilde{x}}$ is the frequency of the median class &
 $n = \sum f_i$.
- **The median class is the class which include**
 $\left(\frac{n}{2}\right)^{th}$ *value.*
- **Example:** The following table shows a frequency distribution of scores on a final examination in Introduction to Statistics for 100 students. Then, obtain median and interpret the result.

Cont'd

Score (interval)	No. of Students
40 – 49	3
50 – 59	11
60 – 69	21
70 – 79	43
80 – 89	22

Soln

Properties of the Median

- It is an average of location, not the average of the values in the data set.
- It is more affected by the number of observations than the extreme values.
- Median can be calculated even in the case of open-ended interval.

4. Quantiles

- The median gives us a value which divides the data set in to **two equal parts**.
- There are also ***other positional measures*** that divide a given data set into more than two equal parts.
- These measures are collectively known as ***quantiles***.
- Quantiles include **quartiles, deciles & percentiles**.

Cont'd

- **Quartiles are some three points that divide the array in to four parts in away each portion contains equal number of observations.**
- **The first, second and third points are called the first (Q_1), second (Q_2) and third (Q_3) quartiles respectively.**
- **25% of the data fall below Q_1 , 50% below Q_2 and 75% below Q_3 &**

$$Q_1 \leq Q_2 \leq Q_3$$

Cont'd

- **Deciles are nine points that divide the array in to ten equal parts.**
- **The first, second, . . . ,ninth deciles are denoted by $D_1, D_2, \dots, D_9$ respectively. 10% of the data fall below D_1 , 20% below $D_2, \dots$, 90% below D_9 and**

$$D_1 \leq D_2 \leq \dots \leq D_9$$

Cont'd

- **Percentiles are ninety nine points that divide the array in to 100 equal parts.**
- **They are denoted by $P_1, P_2, \dots, P_{99}$ and always:**

$$P_1 \leq P_2 \leq \dots \leq P_{99}$$

Cont'd

- †. To calculate quantiles for raw data and data in ungrouped frequency distribution after arranging data in ascending order, we apply the following formulas:

$$Q_i = \left(\frac{i(n+1)}{4} \right)^{th} \text{ value}, i = 1, 2, 3$$

$$D_i = \left(\frac{i(n+1)}{10} \right)^{th} \text{ value}, i = 1, 2, \dots, 9$$

$$P_i = \left(\frac{i(n+1)}{100} \right)^{th} \text{ value}, i = 1, 2, \dots, 99$$

Cont'd

- **Example:** Given the data 420, 430, 435, 438, 441, 449, 490, 500, 510 & 515. Find:

(a) all quartiles

(b) 2nd & 6th deciles

(c) 20th, 65th & 90th percentiles

Soln

Cont'd

2. For data in grouped frequency distribution:

$$Q_i = L_{qi} + \frac{\left(\frac{in}{4} - F_{qi-1}\right)}{f_{qi}} w$$

$$D_i = L_{di} + \frac{\left(\frac{in}{10} - F_{di-1}\right)}{f_{di}} w$$

$$P_i = L_{pi} + \frac{\left(\frac{in}{100} - F_{pi-1}\right)}{f_{pi}} w$$

Cont'd

Where:

- L_{qi}, L_{di}, L_{pi} are the lower class boundaries of the classes containing the concerned quantile points,
- $F_{qi-1}, F_{di-1}, F_{pi-1}$ are the LCF of the class which precedes the class containing the concerned quantile points,
- f_{qi}, f_{di}, f_{pi} are frequencies of classes containing the concerned quantile points &
- w is the class width of a class containing the concerned quantile point.

Cont'd

Note

- Q_i is found in the class containing the $\left(\frac{in}{4}\right)^{th}$ observation.
- D_i is found in the class containing the $\left(\frac{in}{10}\right)^{th}$ observation.
- P_i is found in the class containing the $\left(\frac{in}{100}\right)^{th}$ observation.

Cont'd

- **Example:** Calculate all quartiles, the 7th and 9th deciles, and the 40th and 85th percentiles for the following students score data and interpret the results.

Class Boundaries	Frequency (fi)	LCF
10.5 – 14.5	4	4
14.5 – 18.5	7	11
18.5 – 22.5	8	19
22.5 – 26.5	10	29
26.5 – 30.5	12	41
30.5 – 34.5	7	48
34.5 – 38.5	8	56

Soln

2. Measures of Dispersion

- **Variation or dispersion may be defined as the extent of scatteredness of values in the data set around the measures of central tendency.**
- **Thus, a measure of dispersion tells us the extent to which the values of a variable vary about the measure of central tendency.**

Objectives of Measures of Dispersion

- **To have an idea about the reliability of the measures of central tendency**
- **To compare two or more sets of data with regard to their variability**
- **To provide information about the structure of the data**
- **To pave way to the use of other advanced statistical measures**

Types of Measures of Dispersion

1. **Absolute Measures of Variation:** A measure of variation is said to be an absolute form when it shows the actual amount of variation of an item from a measure of central tendency and are expressed in concrete units in which the data have been expressed.
2. **Relative Measures of Variation:** A relative measure of variation is the quotient obtained by dividing the absolute measure by a quantity in respect to which absolute deviation has been computed. It is a pure number and used for making comparisons between different distributions.

Cont'd

Absolute Measures	Relative Measures
Range	Coefficient of Range
Quartile Deviation	Coefficient of Quartile Deviation
Mean Deviation	Coefficient of Mean Deviation
Variance	Coefficient of Variation
Standard Deviation	Standard Scores

1. Range & Relative Range

- Range is the simplest and crudest/rough measure of dispersion.
- It is defined as the difference between the largest and the smallest values in the data.

For Raw Data:

$$R = L - S$$

$$CR = \frac{L - S}{L + S}$$

Cont'd

■ **For Grouped Data:**

$$R = UCL_{last} - LCL_{first}$$

$$CR = \frac{UCL_{last} - LCL_{first}}{UCL_{last} + LCL_{first}}$$

Where:

R = Range

CR = Coefficient of Range

L = Largest Value

S = Smallest Value

2. Quartile Deviation & Coefficient of Quartile Deviation

- Quartile deviation is sometimes known as Semi-Interquartile Range (SIR). The interquartile Range is $Q_3 - Q_1$. Thus:

$$QD = \frac{Q_3 - Q_1}{2}$$
$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Where:

QD = Quartile Deviation

CQD = Coefficient of Quartile Deviation

3. Mean Deviation & Coefficient of Mean Deviation

- Mean deviation is the arithmetic mean of the absolute values of the deviation from some measures of central tendency usually the mean and the median of a distribution.
- Hence we have mean deviation about the mean $MD(\bar{x})$ and mean deviation about the median $MD(\tilde{x})$.

Cont'd

For Raw Data:

$$MD(\bar{x}) = \frac{\sum |x_i - \bar{x}|}{n} \&$$

$$MD(\tilde{x}) = \frac{\sum |x_i - \tilde{x}|}{n}$$

Cont'd

■ For Grouped Data:

$$MD(\bar{x}) = \frac{\sum f_i |m_i - \bar{x}|}{\sum f_i} \text{ \& }$$

$$MD(\tilde{x}) = \frac{\sum f_i |m_i - \tilde{x}|}{\sum f_i}$$

The coefficients of mean deviations are:

$$CMD(\bar{x}) = \frac{MD(\bar{x})}{\bar{x}} \text{ \& } CMD(\tilde{x}) = \frac{MD(\tilde{x})}{\tilde{x}}$$

Cont'd

- **Example 1:** Consider a sample with data values of **20, 25, 30, 25, 35, 24, 38 & 20**. Compute the range, coefficient of range, quartile deviation, coefficient of quartile deviation, mean deviation about mean, mean deviation about median, coefficient of mean deviation about mean and coefficient of mean deviation about median.

Soln

Cont'd

- **Example 2:** Calculate the QD , CQD , $MD(\bar{x})$, $MD(\tilde{x})$, $CMD(\bar{x})$ & $CMD(\tilde{x})$ for the following frequency distribution.

CL	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
f_i	8	10	22	35	15	10

Soln

4. Variance & Standard Deviation

- Variance and standard deviation are the most superior and widely used measures of dispersions and both measure the average dispersion of the observations around the mean.
- For a *population* containing N elements, the population standard deviation is denoted by the Greek letter σ (sigma) and hence the population variance is denoted by σ^2 .

Cont'd

■ **For Raw Data:** $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$ & $\sigma =$

$$\sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

For Grouped Data: $\sigma^2 = \frac{\sum f_i (m_i - \mu)^2}{N}$ & $\sigma =$

$$\sqrt{\frac{\sum f_i (m_i - \mu)^2}{N}}$$

Cont'd

- For a *sample* of n elements, the sample variance and standard deviation denoted by S^2 and S , respectively, are calculated as using the formula:

For Raw Data: $S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$ & $S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

For Grouped Data: $S^2 = \frac{\sum f_i (m_i - \bar{x})^2}{\sum f_i - 1}$ & $S = \sqrt{\frac{\sum f_i (m_i - \bar{x})^2}{\sum f_i - 1}}$

Cont'd

- **Example 1:** Consider a sample with data values of **10, 20, 12, 17 & 16**. Compute the variance and standard deviation.

Soln

- **Example 2:** Calculate the variance and standard deviation for the following frequency distribution.

CL	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
fi	8	10	22	35	15	10

Soln

5. Coefficient of Variation

- **Coefficient of variation is a relative measure of standard deviation.**
- **It is the ratio of the standard deviation to the mean and expressed as percent.**
- **Hence, it is a unitless measure of variation and also takes into account the size of the means of the distributions.**

Cont'd

For Population: $CV = \frac{\sigma}{\mu} \times 100\%$

For Sample Data: $CV = \frac{S}{\bar{x}} \times 100\%$

- The distribution having less CV is said to be less variable or more consistent or more uniform.
- For field experiments, CV , is generally reported.
- If it is small, it indicates more reliability of experimental findings.

Cont'd

- **Example 1:** Compare the variability of the following two sample data sets using standard deviation and coefficient of variation.

A : 2 Meters, 4 Meters, 6 Meters

B : 1000 Liters, 800 Liters, 900 Liters

Soln

Cont'd

- **Example 2:** The average IQ of statistics students is 110 with standard deviation 5 and the average IQ of mathematics students is 106 with standard deviation 4. Which class is less variable in terms of IQ?

Soln



Chapter 4

Probability & Probability Distributions

What is Probability

- **It is a numerical description of chance of occurrence of a given phenomena under certain condition.**
- **Probability theory plays a central role in statistics.**
- **After all, statistical analysis is applied to a collection of data in order to discover something about the underlying events.**

Definitions of Some Basic Concepts

1. *Experiment (E)*: is any statistical process that can be repeated several times and in any trial of which the outcome is unpredictable.

Examples:

- Tossing a coin only once, $S = \{\text{Head (H), Tail (T)}\}$
- Tossing a coin two times, $S = \{HH, HT, TH, TT\}$
- Rolling a die, $S = \{1, 2, 3, 4, 5, 6\}$
- Selecting an item from a production lot, $S = \{\text{Defective, Non-defective}\}$
- Introducing a new product, $S = \{\text{Success, Failure}\}$

Cont'd

2. **Sample Space (S):** is a set consisting all possible outcomes of a given experiment, .
3. **Event:** is an outcome or a set of outcomes (having some common characteristics) of an experiment.
 - **Simple Event (Elementary Event):** is an event consisting a single outcome. The elementary events are the building blocks (or atoms) of a probability model.
 - They are the events that cannot be decomposed further into smaller sets of events.
 - **Compound Event:** is an event consisting two or more outcomes.

Cont'd

4. ***Independent Event:*** two or more events are independent if the occurrence of one event has no effect on the probability of occurrence of the other.
5. ***Mutually Exclusive Events:*** two or more events are mutually exclusive, if they have no outcome in common. They cannot occur together simultaneously.
6. ***Complementary Event:*** Two mutually exclusive events are complementary if there are no common elements between themselves and both of them contain all possible outcomes.

Counting Rules

- **Counting techniques are mathematical models which are used to determine the number of possible ways of arranging or ordering objects.**
- **They are used to find a solution to fix the size of the sample space that is extremely large.**
- **To count possible outcomes of a sample space or/and an event we use the following counting techniques:**

Cont'd

- †. **Addition Rule:** states that if a task can be done (accomplished) by any of the k procedures, where i^{th} procedures has n_i alternatives, then the total number of ways of doing the task is given by:

$$n_1 + n_2 + \cdots + n_k = \sum_{i=1}^k n_i$$

Cont'd

2. **Multiplication Rule:** states that if a choice consists k steps where the first step can be done in n_1 ways, for each of which second can be done in n_2 ways, for each of those k^{th} steps can be done in n_k ways. Then, the total number of distinct ways to accomplish the task/choice is equal to:

$$n_1 \times n_2 \times \cdots \times n_k = \prod_{i=1}^k n_i$$

Cont'd

- **Example :** Suppose a cafeteria provides 5 kinds of cake which it serves with tea, coffee, milk & coca cola. Then, in how many different ways can you order your breakfast of cake with a drink?

Soln

Given

$$n_1 = 5 \text{ \& } n_2 = 4$$
$$n_1 \times n_2 = 5 \times 4 = 20$$

Cont'd

3. **Permutation:** is arrangement of objects with attention to order of appearance.

Rule 1: The number of permutations of n distinct objects taking all together is:

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times (1)$$

Example : In how many different ways can 3 persons sleep in a bed?

Soln

$$n! = 3! = 3 \times 2 \times 1 = 6 \text{ ways}$$

Cont'd

Rule 2: Given n distinct objects, the number of permutations of r objects taken from n objects is denoted by nPr and given by:

$$nPr = \frac{n!}{(n-r)!}, r \leq n$$

- **Example 1:** In how many ways can 10 people be seated on a bench if only 4 seats are available?

Soln

Cont'd

- **Example 2:** How many 5 letter permutations can be formed from the letters in the word **DISCOVER**?

Soln

Rule 3: Given n objects in which n_1 are alike, n_2 are alike, ..., n_r are alike is given by:

$$\frac{n!}{n_1! \times n_2! \times \cdots \times n_r!}$$

Cont'd

- **Example:** How many different permutations can be made from the letters in the word: **STATISTICS**

Soln

4. **Combination:** A set of n distinct objects considered without regard to the orders of appearance is called combination. For example, *abc*, *bac*, *acb*, *cab*, *cba* are six different permutations but they are the same combination.

Cont'd

■ **Rule 1:** The number of ways of selecting r objects from n distinct objects is called combination of r objects from n objects denoted by nCr or $\binom{n}{r}$ and given by:

$$nCr = \binom{n}{r} = \frac{n!}{(n-r)! \times r!}; r \leq n$$

- **Example:** In how many ways can student choose 3 books from a list of 12 different books?

Soln

Cont'd

Rule 2: If the selection has k steps, by selecting r_1 of n_1 objects, r_2 of n_2 , ..., r_k of n_k objects, then the total number of ways of doing this selection is equal to:

$$\binom{n_1}{r_1} \times \binom{n_2}{r_2} \times \cdots \times \binom{n_k}{r_k}$$

- **Example:** Out of 5 male workers and 7 female workers of some factory a committee consisting 2 male and 3 female workers to be formed. In how many ways can this be done if:

Cont'd

- (a) all workers are eligible**
- (b) one particular female must be a member**
- (c) two particular male workers cannot be members for some reason**

Soln

- **The difference between permutation and combination is that in combination the order of objects being selected (arranged) is not important, but order matters in permutation.**

Approaches in Probability Assignment

1. **The Classical Approach (also called Mathematical Approach):** Suppose there are N possible outcomes in the sample space S of an experiment. Out of these N outcomes, only n are favorable to the event E , then the probability that the event E will occur is:

$$\begin{aligned} P(E) &= \frac{\text{No of favorable outcomes to } E}{\text{Total no of outcomes}} \\ &= \frac{n(E)}{n(S)} = \frac{n}{N} \end{aligned}$$

Cont'd

Example : Consider an experiment of rolling a die. Then, what is the probability that:

- a) Odd numbers occur?
- b) Number 4 occurs?

Soln

2. **The Empirical Approach (also called Frequency Approach):** It is based on a relative frequency. Given a frequency distribution, the probability of an event being in a given class is:

Cont'd

- $$P(E) = \frac{f_i}{\sum f_i}$$

3. **Subjective Approach:** calculates probability based on an educated guess or experience or evaluation of a problem. For example a physician might say that on the basis of his/her diagnosis, there is a 30% chance the patient will need an operation.

Some Probability Rules/Axioms

- $0 \leq P(E) \leq 1$
- **If 'S' is sample space, $P(S) = 1$**
- **If A & B are mutually exclusive events, then**
 $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$
- **If $A_1, A_2, \dots, A_n$ are pairwise mutually exclusive events, then:**

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Cont'd

- $P(A \cup A^c) = P(A) + P(A^c)$
- $P(\emptyset) = 0$
- **Using the above axioms, it can be shown that for any two events A and B,**
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 1: A political candidate runs for two offices, A and B. The candidate assesses the chance of winning A at 0.3 of winning B at 0.4 and of winning both at 0.1. What is the probability that the candidate wins office A or B?

Soln

Cont'd

- **Example 2:** A student is taking two courses marketing and statistics. The probability that the student will pass marketing course is 0.6 and the probability of passing the statistics course is 0.7 and the probability of passing both is 0.5. What is the probability of passing at least one of the course?

Soln

Conditional Probability

- Often the probability of an event is influenced by whether a related event has occurred.
- Suppose we have an event 'A' with probability $P(A)$.
- If we obtain new information and learn that a related event denoted by 'B' has occurred; we want to take the advantage of this information in calculating a new probability for event 'A'.

Cont'd

- This new probability of event 'A' is written **$P(A/B)$** .
- The notation is used to denote the fact that we are considering the probability of event 'A' given the condition that event 'B' has occurred.
- Hence the notation $P(A/B)$ is read ***“The probability of A given B”***.

Cont'd

- The fact that conditional probabilities can be computed as the ratio of a **joint probability** to a **marginal probability** provides the following general formula for conditional probability calculation for two events A & B:

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \text{ \& } P(B/A) = \frac{P(B \cap A)}{P(A)}$$

Cont'd

- **Example 1:** Consider the situation of the promotion status of male and female officers of a certain hypothetical metropolitan. The police force consists of 1200 officers, 960 men and 240 women. Over the past two years, 324 officers of the police force have been awarded promotions. The specific breakdown of promotion for male and female is shown in Table 4.1 below:

Cont'd

- **Table 4.1: Promotion status of police officers over the past two years**

	Men	Women	Total
Promoted	288	36	324
Not Promoted	672	204	876
Total	960	240	1200

- **After reviewing the promotion record, a committee of female officers raised discrimination issue on the basis that 288 male officers have received promotions but only 36 female officers have received promotion.**

Cont'd

- The police administration argued that the relatively low number of promotion for female officers was not due to discrimination but due to the fact that there are relatively few female officers in the police force. Use conditional probability to analyze the discrimination charge and make decision.

Soln

Let: M = event an officer is a man

W = event an officer is a woman

A = Event an officer is promoted

$\tilde{A}$ = event an officer is not promoted

Cont'd

- Summary of the joint & marginal probabilities is given in the following table 4.2.

	Men (M)	Women (W)	Total
Promoted (P)	0.24	0.03	0.27
Not Promoted ($\dot{P}$)	0.56	0.17	0.73
Total	0.80	0.20	1.00

Required

Find: $P(A/M)$ & $P(A/W)$

**Joint
Probabilities**

**Marginal
Probabilities**

Cont'd

- **Example 2:** In a firm, 20% of the employees have accounting backgrounds; while 5 percent of employees are executives and have accounting backgrounds. If an employee has an accounting background, what is the probability that the employee is an executive?

Soln

Independent Events

- **Two events are independent if; $P(A/B) = P(A)$ & $P(B/A) = P(B)$.**
- **Whereas the addition law of probability is used to compute the probability of a union of two events, the multiplication law is used to compute the probability of an intersection of two events.**
- **The multiplication law is based on the definition of conditional probability.**

Cont'd

- **The Multiplication law states that:**

$$P(A \cap B) = P(A) \times P(B/A)$$

or

$$P(A \cap B) = P(B) \times P(A/B)$$

- **Example 1:** Suppose that there are 10 rolls of a film in a box of which 3 are defective. Two rolls are to be selected one after the other *without replacement*. What is the probability of selecting a defective roll followed by another defective roll?

Soln

Cont'd

- If A & B are two independent events the intersection of the two is given by special multiplication rule as follows:

$$P(A \cap B) \text{ or } P(A \& B) = P(A) \times P(B)$$

- To put it in another way, two events are independent if the outcome of the second event doesn't depend on the outcome of the first event.
- **Example 2:** Suppose two coins are tossed, what is the probability that both will land tail up?

Soln

Baye's Theorem

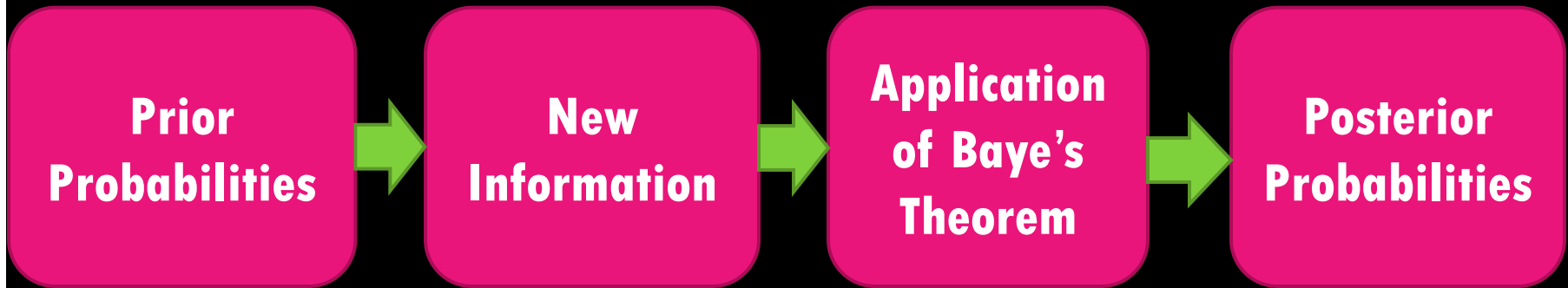
- **Revising probability when new information is obtained is an important phase of probability analysis.**
- **Often, we begin the analysis with initial or prior probability estimate for specific events of interest.**
- **Then, from sources such as a sample, a special report or a product test, we obtain additional information about the events.**

Cont'd

- Given this new information, we update the **prior probability** (initial estimates of the probabilities of events) values by calculating revised probabilities, referred to as **posterior probabilities** (revised probabilities of events based on additional information).
- *Baye's theorem* provides means for making these probability calculations.

Cont'd

- The probability revision using Baye's theorem can be shown as follows:



Cont'd

- **Example:** Consider a manufacturing firm that receives shipments of parts from two different suppliers. Let A_1 denote the event that a part is from supplier 1 and A_2 denote the event that a part is from supplier 2. Currently, 65% of the parts purchased by the company are from supplier 1 and the remaining 35% are from supplier 2. Hence, if a part is selected at random we would assign the prior probabilities $P(A_1) = .65$ and $p(A_2) = .35$

Cont'd

- The quality of the purchased parts varies with the source of supply. Historical data suggest that the quality ratings of the two suppliers are as follows:

	%tage of Good Parts	%tage of Bad Parts
Supplier 1	98	2
Supplier 2	95	5

Cont'd

- Suppose now that the parts from the two suppliers are used in the firm's manufacturing process and that a machine breaks down because it attempts to process a bad part. *Given the information that the part is bad, what is the probability that it came from supplier 1 and what is the probability that it came from supplier 2?*

Soln:

Cont'd

- Let '**A₁**' represent supplier 1, '**A₂**' represent supplier 2, '**G**' represent good parts and '**B**' represent bad parts.
- From the law of conditional probability, we know that:

$$P\left(A_1/B\right) = \frac{P(A_1 \cap B)}{P(B)}$$

- Referring to multiplication rule:

$$P(A_1 \cap B) = P(A_1)P(B/A_1)$$

Cont'd

- To find $P(B)$, we note that event 'B' can occur in only two ways: $(A_1 \cap B)$ and $(A_2 \cap B)$.

Therefore, we have:

$$P(B) = P(A_1)P\left(\frac{B}{A_1}\right) + P(A_2)P\left(\frac{B}{A_2}\right)$$

- So, the substituting the above formulas we can obtain the Baye's theorem for the case of two events.

Cont'd

- **Baye's Theorem (Two Event Case):**

$$P(A_1/B) = \frac{P(A_1)P(B/A_1)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2)} \&$$

$$P(A_2/B) = \frac{P(A_2)P(B/A_2)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2)}$$

Cont'd

$$\text{So, } P(A_1/B) = \frac{P(A_1)P(B/A_1)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2)}$$

$$= \frac{(.65)(.02)}{(.65)(.02) + (.35)(.05)} = .43$$

$$P(A_2/B) = \frac{P(A_2)P(B/A_2)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2)}$$

$$= \frac{(.35)(.05)}{(.65)(.02) + (.35)(.05)} = .57$$

Cont'd

- **Baye's Theorem ('n' Events case):**

$$P(A_i/B) = \frac{P(A_i)P(B/A_i)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2) \dots + P(A_n)P(B/A_n)}$$

The Tabular Approach to Baye's Theorem

- A tabular approach is helpful in conducting the Baye's' theorem calculations.
- The tabular computations are done in the following steps:

Step 1. Prepare the following three columns:

- Column 1: The mutually exclusive events for which posterior probabilities are desired.
- Column 2: The prior probabilities for the events.
- Column 3: The conditional probabilities of the new information given each event.

Cont'd

- **Step 2.** In column 4, compute the joint probabilities for each event and the new information B by using the multiplication law.
- **Step 3.** Sum the joint probabilities in column 4.
- **Step 4.** In column 5, compute the posterior probabilities using the basic relationship of conditional probability

Cont'd

Summary Of Baye's Theorem Calculations For The Two Supplier Problem:

(1)	(2)	(3)	(4)	(5)
Events	Prior Probabilities	Conditional Probabilities	Joint Probabilities	Posterior Probabilities
A_i	$P(A_i)$	$P(B/A_i)$	$P(A_i \cap B)$	$P(A_i/B)$
A1	.65	.02	.0130	$.0130/.0305$ = .43
A2	.35	.05	.0175	$.0175/.0305$ = .57
	1.00		$P(B) = .0305$	1.00

Probability Distributions

- Consider the following illustration:
- **Example:** Consider the experiment of tossing a coin twice.
 $S = \{HH, HT, TH, TT\}$
- Let ' X ' be number of heads. Thus, another sample space with respect to X (also called the range space of X) is: $R_x = \{0, 1, 2\}$
- A function ' X ' which assigns a real numbers to all possible values of a sample space is called a **random variable**.
- A random variable is a variable that has a single numerical value (determined by chance) for each outcome of a procedure.

Cont'd

- A random variable can be classified as being either ***discrete or continuous*** depending on the numerical values it assumes.
- A ***discrete random variable*** has either a finite number of values or a countable number of values; that is, they result from counting process.
- A ***continuous random variable*** has infinitely many values, and those values can be associated with measurements on a continuous scale in such a way that there are no gaps or interruptions.

1. Common Discrete Probability Distributions

A. Binomial Probability Distribution

- The binomial probability distribution is a discrete probability distribution that provides many applications.
- It is associated with a multiple-step experiment that we call the **binomial experiment**.
- A binomial experiment exhibits the following four properties:
 1. The procedure has a fixed number of trials.

Cont'd

2. The trials are independent (the outcome of any individual trial does not affect the probabilities in the other trials).
3. The outcome of each trial must be classifiable into one of two possible categories (*success or failure*).
4. The probability of a success, denoted by *p*, does not change from trial to trial.

Cont'd

- If a procedure satisfies these four requirements, the distribution of the random variable (X) is called a binomial probability distribution (or binomial distribution).
- To calculate the probabilities we use the following formula:

$$P(X = x) = \binom{n}{x} p^x q^{n-x} \text{ for } x = 0, 1, 2, \dots, n$$

Where;

Cont'd

x = the number of successes

p = the probability of a success on one trial

**q = the probability of failure on one trial
($q = 1 - p$)**

n = the number of trials

$p(x)$ = the probability of ' x ' successes in ' n ' trials.

- Expected value and variance of binomially distributed random variable $[X, \text{Bin}(n, p)]$ can be obtained using the following formulas:

Cont'd

- $E(X) = \mu = np$
- $Var(X) = \sigma^2 = np(1 - p) = npq$
- $SD(X) = \sigma = \sqrt{np(1 - p)} = \sqrt{npq}$
- **Example:** A university found that 20% of its students withdraw without completing the introductory statistics course. Assume that 20 students registered for the course. Compute the probability that:
 - a) Exactly four will withdraw
 - b) At most two will withdraw

Cont'd

- c) **More than three will withdraw**
- d) **Calculate mean & standard deviation of the withdrawals.**

Soln

- a) **0.2182**
- b) **0.2061**
- c) **0.5886**
- d) **4 & 1.79 respectively**

Cont'd

Example 2: A shoe store's records show that 30% of customers making a purchase use a credit card to make payment. Suppose that this morning, 15 customers purchased shoes from the store, then:

- a) What is the probability that at least 12 of the customers used a credit card?
- b) What is the probability that at least 3 customers, but not more than 6, used a credit card?

Cont'd

- c) What is the expected number of customers who used a credit card? What is the variance?
- d) Find the probability that exactly 14 customers did not use a credit card
- e) Find the probability that at least 9 customers did not use a credit card.

Soln

- | | |
|----------------------------|-----------|
| a) 0.0001 | d) 0.0306 |
| b) 0.7421 | e) 0.8689 |
| c) 4.5 & 3.15 respectively | |

B. Poisson Probability Distribution

- This is a discrete random variable that is often useful *in estimating the number of occurrences over a specified interval of time or space.*
- **For example:** the random variable of interest might be the number of arrivals at a car wash in one hour, the number of repairs needed in 10 miles of highway, or the number of leaks in 100 miles of pipeline.

Properties of Poisson Experiment

1. The probability of an occurrence is the same for any two intervals of equal length.
2. The occurrence or nonoccurrence in any interval is independent of the occurrence or nonoccurrence in any other interval.
- The Poisson probability function is defined by the following equation:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Where, μ = expected value or mean number of occurrences in an interval which is equal to its variance.

Cont'd

Example 1: A student finds that the average number of amoeba in 10 ml of pond water is 4. Find the probability that in 10 ml of water from that pond there are:

- (a) exactly 5 amoeba
- (b) no amoeba
- (c) fewer than 3 amoeba

Soln = 0.1563, 0.0183 & 0.2381 respectively.

Example 2: If X holds Poisson distribution with standard deviation of 3, then find $P(X = 3)$.

Soln = 0.015

Cont'd

Example 3: A parking lot operator has observed that cars arrive randomly at an average rate of 360 cars per hour.

- a) Calculate the probability that only 2 cars will arrive during a specified 1 minute period.
- b) Find the probability that at least four cars will arrive during a specified 1 minute period.

Soln

- a) **0.0446**
- b) **0.8488**

2. Common Continuous Probability Distribution

Normal Probability Distribution

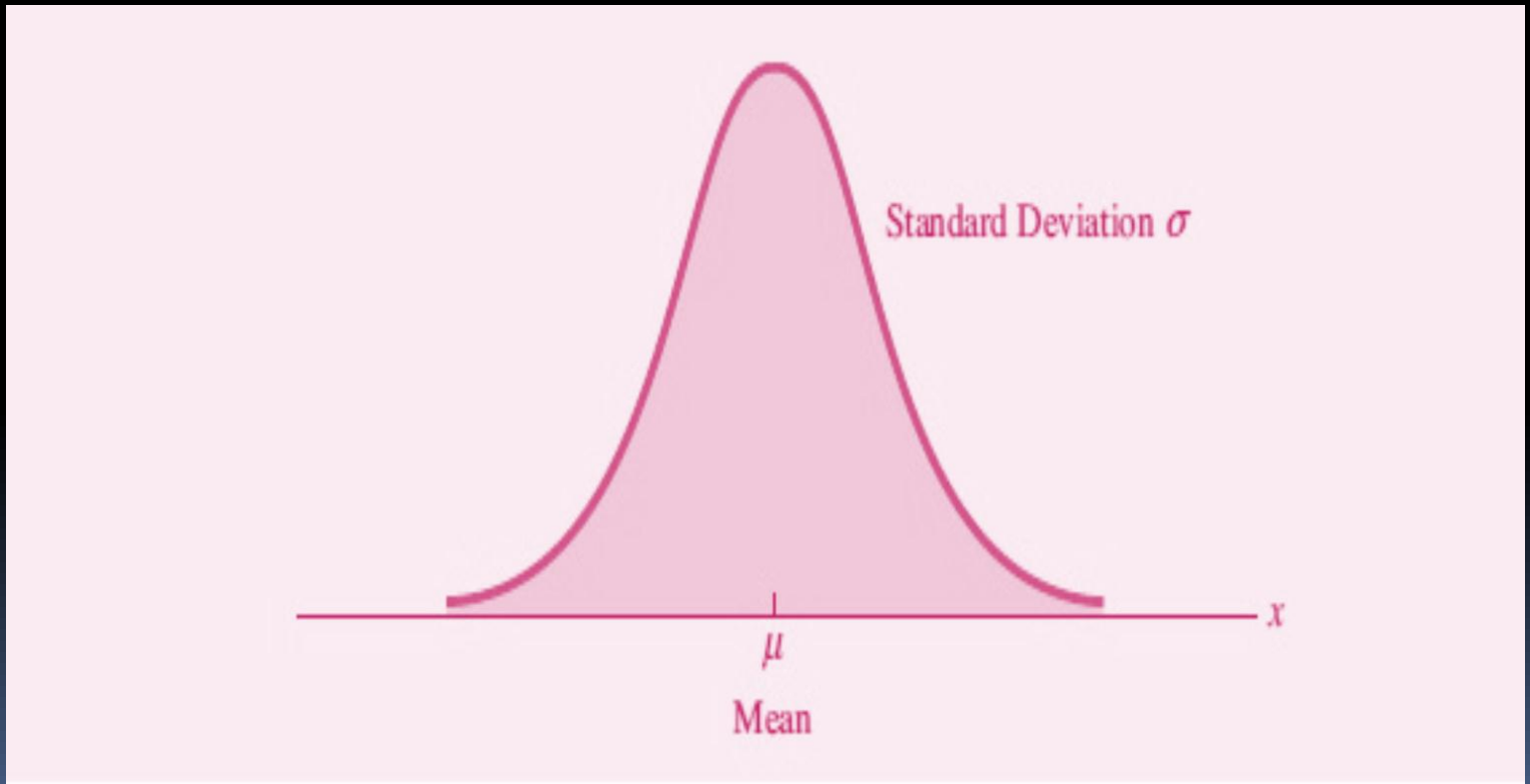
- **The most important probability distribution for describing a continuous random variable is the normal probability distribution.**
- **The normal distribution has been used in a wide variety of practical applications in which the random variables are heights and weights of people, test scores, scientific measurements, amounts of rainfall and other similar values.**

Cont'd

- **It is also widely used in statistical inference.**
- **In such applications, the normal probability distribution provides a description of the likely results obtained through sampling.**

Normal Curve

- The form or shape of the normal distribution is illustrated by the bell-shaped normal curve in the following figure:



Cont'd

- The probability density function (pdf) that defines the bell-shaped curve of the normal distribution as follows:
- If a random variable $X \sim N(\mu, \sigma^2)$, its probability density function (pdf) is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(x-\mu)^2}}{2\sigma^2} \text{ where; } -\infty < x < \infty$$

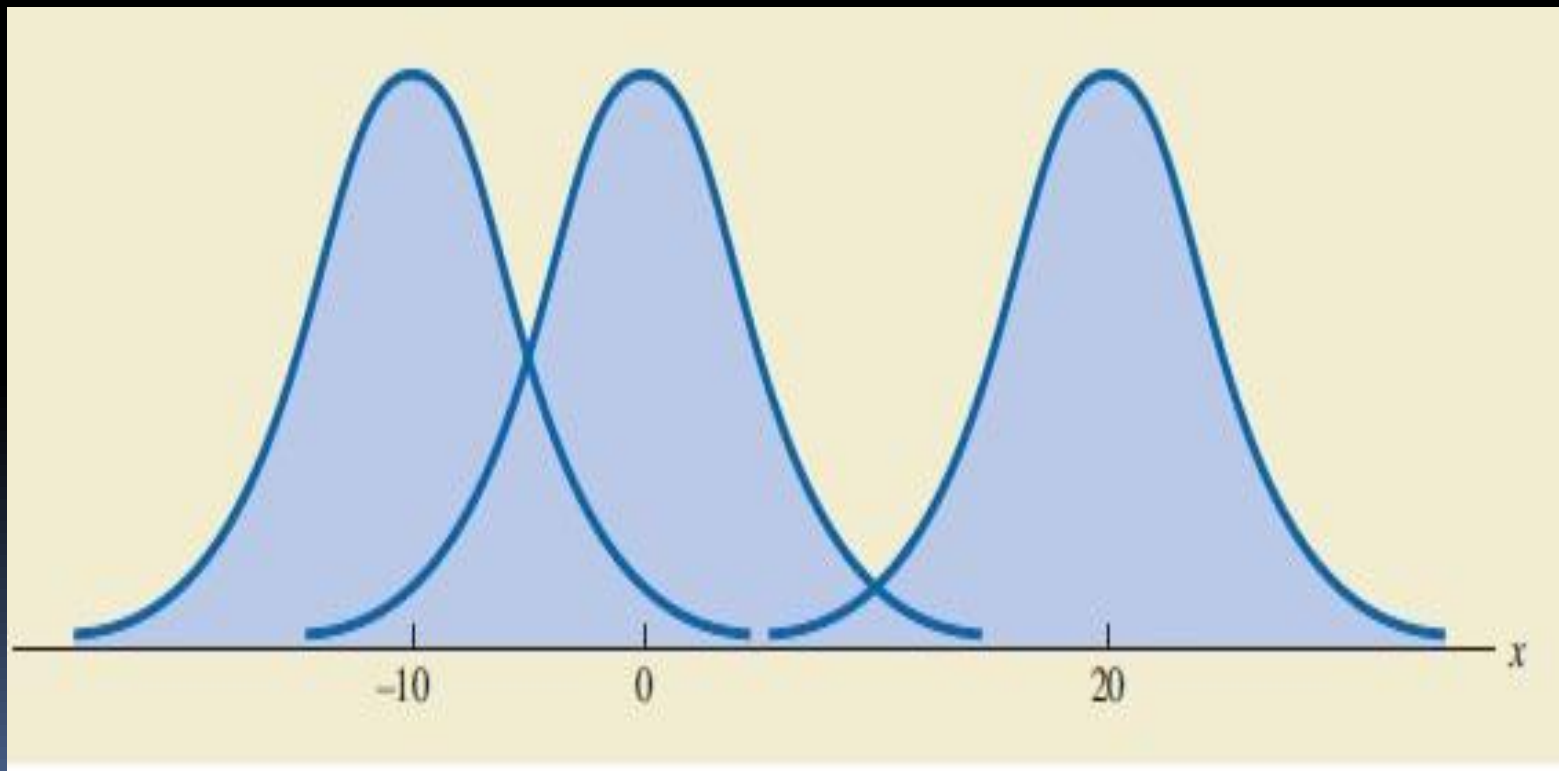
- The normal curve has two parameters, μ (mean) & σ (standard deviation).
- They determine the location and shape of the normal distribution

Properties of Normal Probability Distribution

- The entire family of normal distributions is differentiated by two parameters: the mean μ and the standard deviation σ .
- The highest point on the normal curve is at the mean, which is also the median and mode of the distribution.
- The mean of the distribution can be any numerical value: negative, zero or positive.

Cont'd

- Three normal distributions with the same standard deviation but three different means (-10, 0 & 20) are shown here:



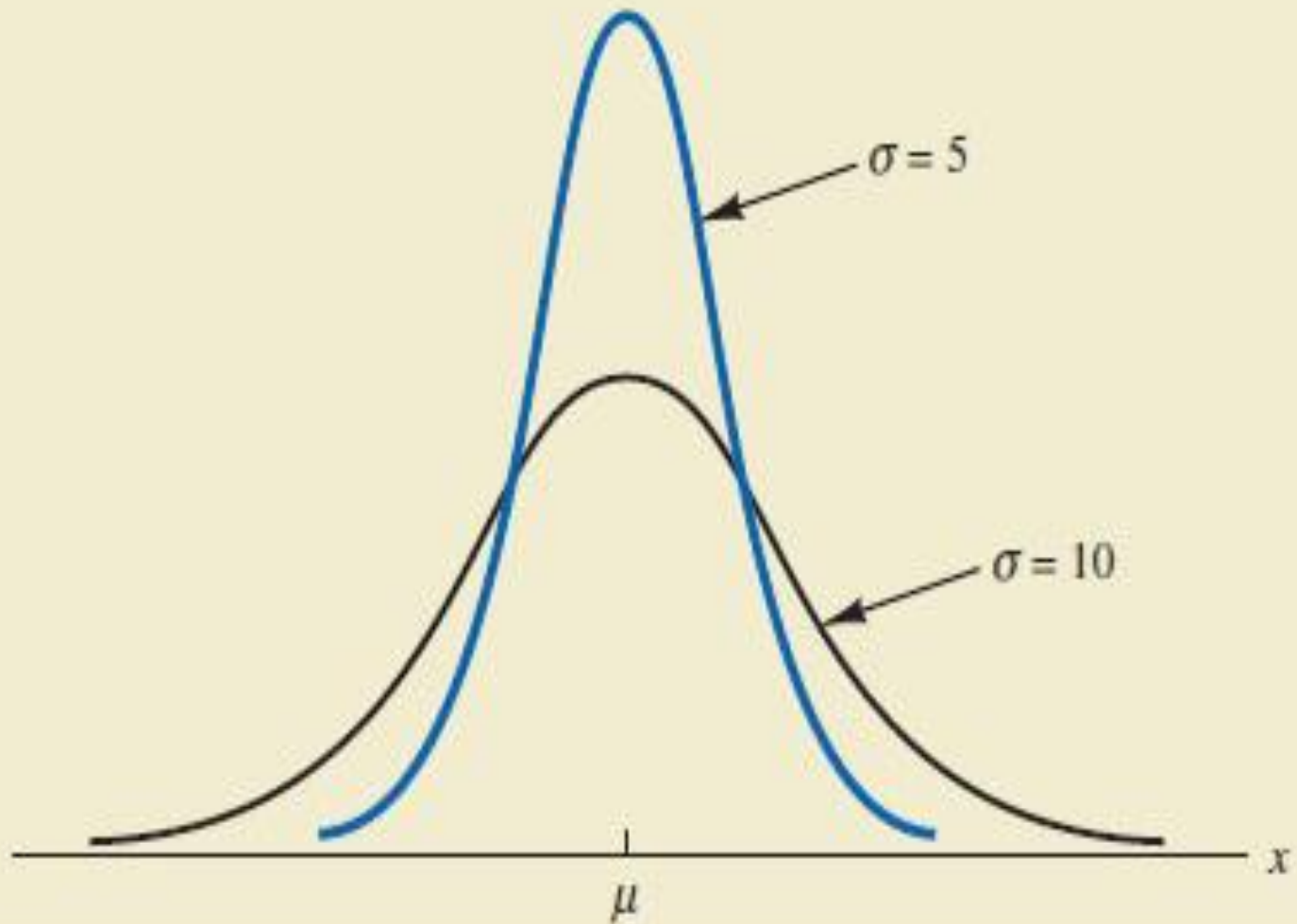
Cont'd

- The **normal distribution is symmetric**, with the shape of the normal curve to the left of the mean a mirror image of the shape of the normal curve to the right of the mean.
- The tails of the normal curve extend to infinity in both directions and theoretically never touch the horizontal axis. Because it is symmetric, the normal distribution is not skewed; its skewness measure is zero.

Cont'd

- **The standard deviation determines how flat and wide the normal curve is.**
- **Larger values of the standard deviation result in wider, flatter curves showing more variability in the data.**
- **Two normal distributions with the same mean but with different standard deviations are shown below:**

Cont'd



Cont'd

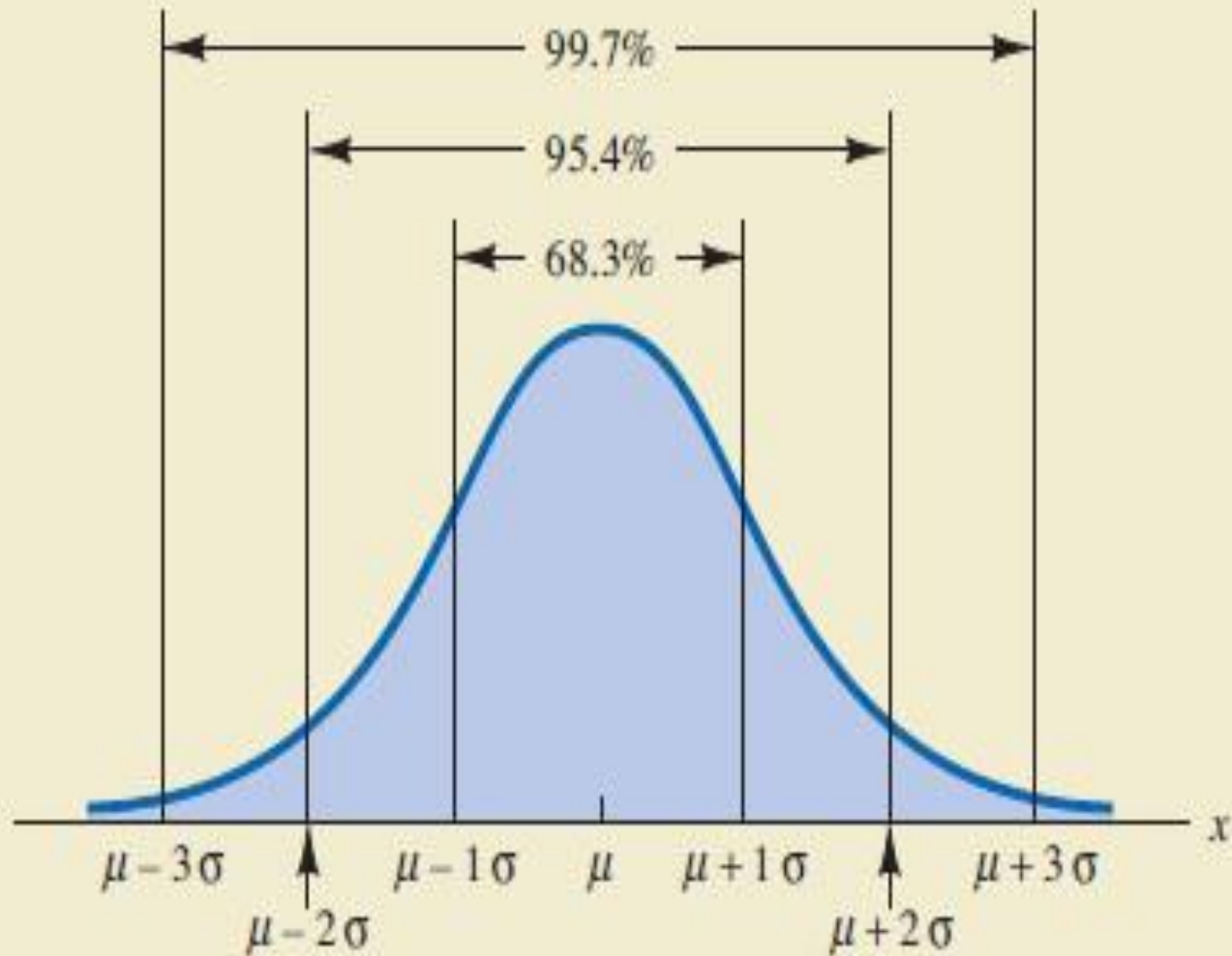
- Probabilities for the normal random variable are given by areas under the normal curve.
- The total area under the curve for the normal distribution is **1**.
- Because the distribution is symmetric, the area under the curve to the left of the mean is **0.50** and the area under the curve to the right of the mean is **0.50**.

Cont'd

The percentage of values in some commonly used intervals are:

- **68.3% of the values of a normal random variable are within plus or minus one standard deviation of its mean.**
- **95.4% of the values of a normal random variable are within plus or minus two standard deviations of its mean.**
- **99.7% of the values of a normal random variable are within plus or minus three standard deviations of its mean.**

Cont'd



Standard Normal Probability Distribution

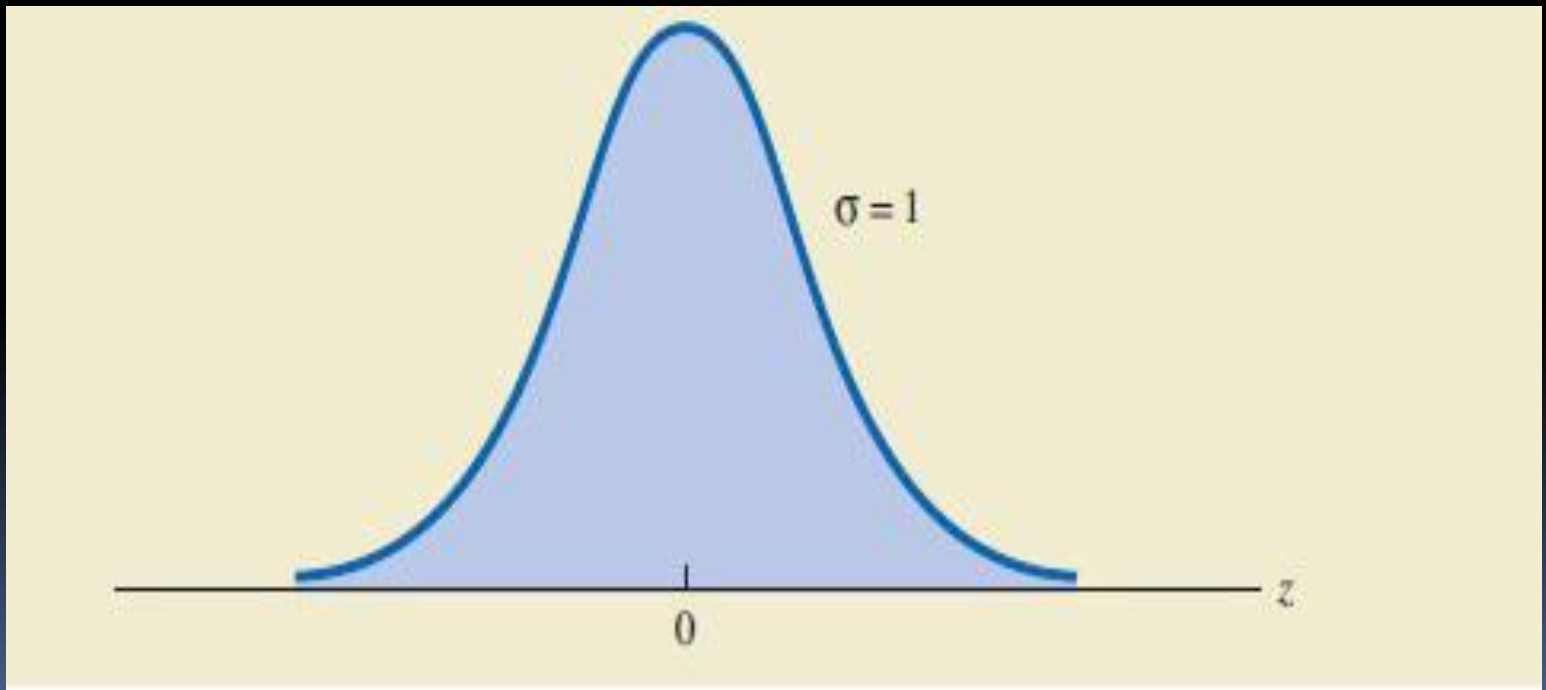
- A random variable that has a normal distribution with a **mean of zero** and a **standard deviation of one** is said to have a standard normal probability distribution.
- The letter '**Z**' is commonly used to designate this particular normal random variable, that is; **$Z \approx N(0, 1)$** .

Cont'd

- when we have a normal distribution with any mean μ and any standard deviation σ , we answer probability questions about the distribution by first converting to the standard normal distribution.
- Then we can use the standard normal probability distribution table and the appropriate 'Z' values to find the desired probabilities.
- Thus, we can convert a given value to 'Z' value using the following formula:

Cont'd

- $$Z = \frac{x - \mu}{\sigma}$$
- **The standard normal probability distribution is shown graphically as follows:**



Cont'd

Example 1: The college boards, which are administered each year to many thousands of high school students, are scored so as to yield a mean of 500 and a standard deviation of 100. These scores are close to being normally distributed. What percentage of the scores can be expected to satisfy each of the following conditions?

- a) Greater than 600
- b) Less than 450
- c) Between 450 & 600

Soln: 0.1587, 0.3085 & 0.5328

Cont'd

Example 2: The weekly incomes of a large group of middle managers are normally distributed with a mean of Br 1000 and standard deviation of Br 100.

- 1. What is the area under the normal curve between Br 1000 & Br 1100?**
- 2. what is the probability of selecting an income between Br 840 & Br 1200?**
- 3. what percent of the middle managers earn an income of Br 1245 or more?**
- 4. what is the area under the normal curve between Br 1,150 & Br 1,250?**

Cont'd

Soln

1. 0.3413

2. 0.9224

3. 0.0071

4. 0.0606

Cont'd

Example 3: Suppose a wholesaler of fertilizer products finds that the annual demand for DAP fertilizer is normally distributed with a mean of 120 tons and standard deviations of 16 tons. If he/she orders only once a year, what quantity should be ordered to ensure that there is only a 5 percent chance of running short?

Soln

- **146.24 tons $\approx$ 147 tons**



THE END!!

THANK YOU SO MUCH!!



ANY QUESTION &/or IDEA?